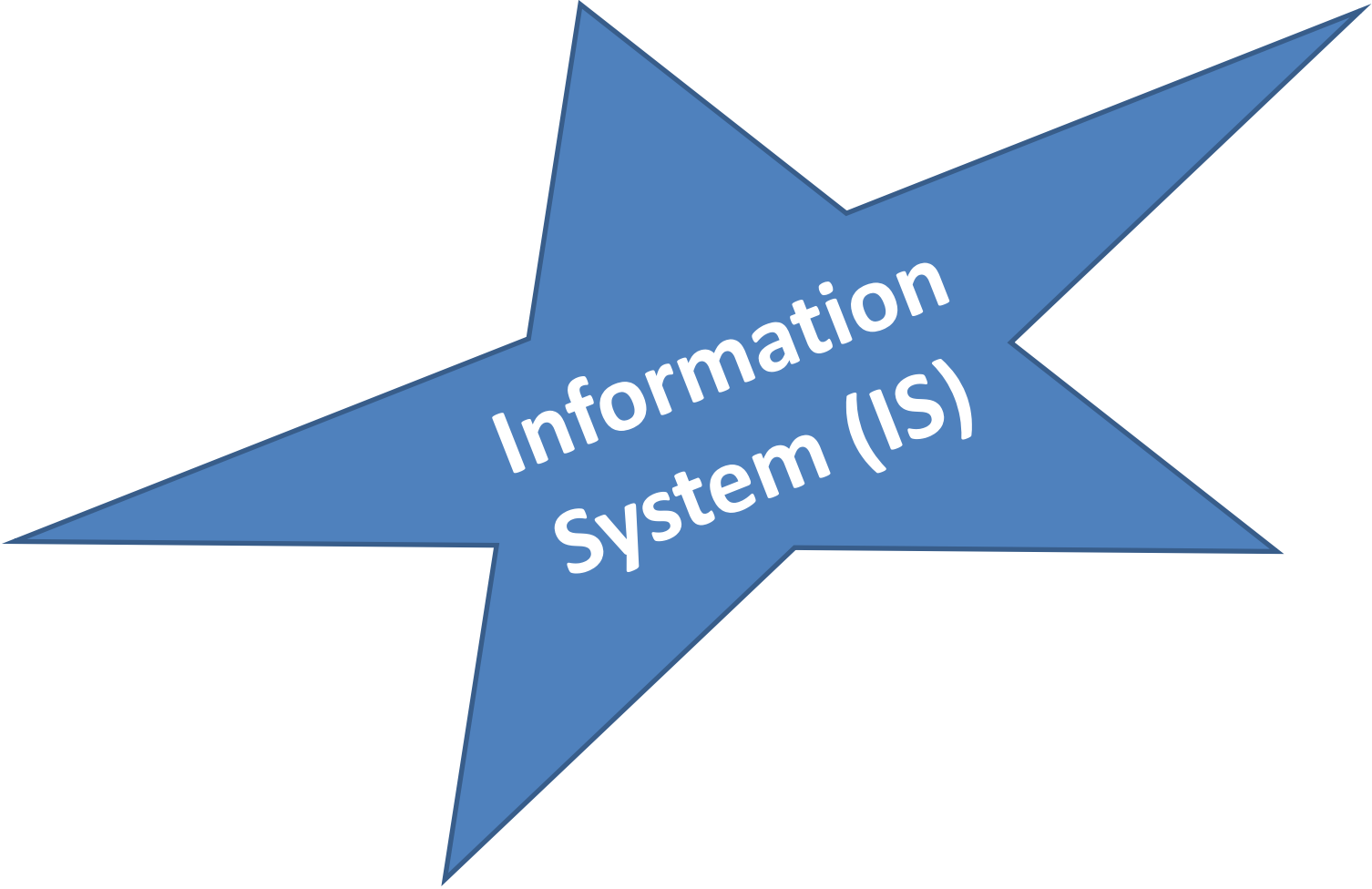


Case Study

Analysis and Design Stages

in

Willow brook School



Information System (IS)

Willow brook School Case Study – Introduction

- Willow brook School is a **small, private school** in the Midwest United States. As an independent systems analyst, you work as an IT consultant, specializing in developing **IT solutions for small businesses**. You have been contacted by the **director, Victoria Owens**, to discuss the possibility of setting up a computer system to **handle some of the school's administrative and financial tasks**. You agree to meet with Victoria and the **principal, Kathy Gilliard** next week to discuss the **school's need for an information system**.

Willow brook School Case Study – Introduction

- **Victoria** explains her role as the **executive director of the school**. She **administers the activities** of the school in accordance with the **mission, vision, and policies** established by the Board of Directors. She **supports the educational staff** and **oversees the financial, payroll, and human resources functions** for the school. **Susan Brown** is the vice principal. She is responsible for the **after-care program**. **Michelle Madrid** is the **administrative assistant** reporting to Victoria. There are **six part-time after-care teachers**, **three for the pre-school/kindergarten** group, and **three for the primary grades**. After-care teachers **report to Susan Brown**.

Willow brook School Case Study – Current IS

- The following **activities** are undertaken by the school, **Admission, Enrolment, Attendance management, Accounts, Transport Management and the Library Database**. The school also supports the local charity by **selling artefacts and toys made by the students** and **maintains an online website** where people can buy and look at the new toys and crafts made by the students, which is **user friendly and has images of the artefacts, toys and crafts**.

Willow brook School Case Study – Current Problem

- **Michelle, the administrative assistant**, in particular is feeling the strain of the increased workload associated with the rapid growth in the student population. The system that she has been using, **of spreadsheets and word processing documents**, worked well when enrollments were smaller, but **now the amount of time spent copying information from one place to another is daunting, and the possibility of errors is increasing. An information system is necessary to cope with the growing administrative workload**

Q1: identify various types of Information Systems (IS) that are currently operating in Willow brook school.

- The type and number of the IS's
 - Admission and Enrolment IS - BS
 - Attendance management IS - BS
 - Accounts System - AFM
 - Transport Management IS – Collaboration IS
 - Library Database IS – TP, KMS, BS
 - Website and Content Management IS - TP, KMS, CRM

Q2: Identify the different components of an information system you identified in Q1

- Hardware – Computers and Peripherals Used
- Software – The **Operating System** the computer runs on, **Microsoft Office products and software** used to maintain admission, enrollment. **Browsers** to access the online website.
- People – School staff and Students, Customers of the website
- Data – **Student** records, **Library** records, **Images of Artifacts** on website
- Procedures (processes) – All **procedure documents** for admission, sale of goods, transport

Q3: part of a **system analyst** role is to examine current IT systems and suggest improvements. *Assuming that you are a **system analyst for Willow Brook School** and you are to examine their current systems, list the problems currently faced.*

- Due to the lack of Integration,
 - work is often **doubled up in collating data** between the Schools different Information systems, and
 - is making the **system more prone to errors (system easily getting errors)** due to the **increasing work load and student intake**.
- So, the problem in the above case study is the **lack of System integration** between the Schools different Information systems.



SDLC Methodology

Q: Recommend the SDLC methodology you would use in developing the Information System for Willowbrook and explain in detail why you have recommended it.

Why we need the IS in this case study?

- **1 . Redundant** work being done in copying pasting
- **2. Excessive time** spent by Michele in **managing the student administrative** system.
- **3. The school has different activities undertaken and does not have a synchronization of data between the activities.** This is a manual task and leading to many errors.

Q: Recommend the SDLC methodology you would use in developing the Information System for Willowbrook and explain in detail why you have recommended it.

What does the new system do?

Though it calls for a **brand-new information system** which seamlessly **integrates all of the schools activity** into one with **reuse and reprocessing of data**, the immediate problem is the growing number of Student enrolment.

Q: Recommend the SDLC methodology you would use in developing the Information System for Willowbrook and explain in detail why you have recommended it.

Why Agile is recommended?

- Hence an **Agile software implementation** would be recommended due to the below
- **Work can be done incrementally** rather than following a waterfall model where the entire requirements are finalized. Using a **waterfall model is not suitable** as the current problem will be faced until the information system is up and running in full swing. **Agile solves this by doing incremental** and therefore using Agile we would first tackle the **Student enrollment** so that the administrative workload decreases.
- Also **Agile gives the school the flexibility** to later bring about **new changes if needed** to the remaining legacy systems and also **provide opportunities to upgrade and enhancements**.
- Agile gives the school the **flexibility to change requirements** in implementing the information system depending on the factors such as **student growth, admin growth and need for a larger database**
- Agile also proves to be **cost effective** as it **solves the immediate problem of the administrative workload** and thereby **reducing costs** in overtime and **increases efficiency**.



Willow Brook School - Background

- **Willow brook School is a small private school** that has retained your services as a systems analyst to assist in the development of a new information system for the school's administrative needs.
- Willow brook school being a small private school is looking to expand and develop into one of the niche school with a unique selling proposition of having few number of students per class.

Willow Brook School – Strength and Weakness

- After an initial conversation with Victoria and Kathy, the school's administrators, they have decided that **pursuing the possibility of an information system is a good idea**. Michelle, the administrative assistant, in particular is feeling the strain of the increased workload associated with the rapid growth in the student population. The system that she has been using, of spreadsheets and word processing documents, worked well when enrollments were smaller, but now the amount of time spent copying information from one place to another is daunting, and the possibility of errors is increasing. **An information system is necessary to cope with the growing administrative workload**. The school management is now proposing to Outsource the School administrative process to **PWC and Co**. **PWC and Co have been in the field of administration system for years and has expertise in doing administrative activities for schools and colleges**. PWC and Co were recently in news for the wrong reasons where in **one of their employees sold confidential data about a college to another college** and thus favoring a potential tender. PWC and Co has however updated their due diligence and security standards post the incident and are slowly gaining back customer confidence.

Willow Brook School – Opportunity

- **Use of technology and providing IPADS to students and eliminate the need of notebooks.** With these ideas in mind it is a tough decision for **Willow brook to decide** whether to **outsource the admin tasks or keep doing them in house and spend on training, maintenance and support and documentation and archiving of information.**

Q1: Write a **Business Case outlining **Background**, current situation, problem, benefits, risks for doing this project and risks of not doing this project to Victoria detailing the need for an integrated information system.**

- **Background:** Willow brook school is a small private school looking to develop a new information system for the schools *administrative needs*. Willow brook school being a small private school is looking to expand and develop into **one of the niche school with a unique selling proposition of having few number of students per class** .

Q1: Write a **Business Case outlining Background, **current situation**, problem, benefits, risks for doing this project and risks of not doing this project to Victoria detailing the need for an integrated information system.**

Current Situation: The school is currently experiencing a high number of student intakes, which **has resulted in increased workload and increased administrative activities**. The increased work load has resulted in errors and lots of manual processing and with the school slated to grow in numbers, there is a need for a management information system.

Q1: Write a **Business Case outlining Background, current situation, **problem**, benefits, risks for doing this project and risks of not doing this project to Victoria detailing the need for an integrated information system.**

Current Problem:

- The system that she has been using, of **spreadsheets and word processing documents**, worked well **when enrolments were smaller**, but now the amount of **time spent copying information** from one place to another is daunting, and the **possibility of errors is increasing**.
- An **information system** is necessary to **cope with the growing administrative workload**.
- The **need for the information system arises from the following situation and problems**.
 - The strain on the administration staff on the **increased workload**
 - **Time spent** in copying information between one system to the another is high.
 - **Number of errors** is increasing
 - **Cost factor** is increasing as the **admin staff are being paid for overtime**. Maintenance of multiple records in different system is increasing cost

Q1: Write a **Business Case outlining Background, current situation, problem, **benefits**, risks for doing this project and risks of not doing this project to Victoria detailing the need for an integrated information system.**

Advantages of the IS: The solution to implement the information system will address the following key business issues

1. **Manual Workload** on the administrative staff
2. **Reduction of errors** through **automation** of data
3. **Reusability of data** once entered in the information system **for use across multiple departments.**
4. **Data security** and scope **for data archiving**

Q1: Write a **Business Case outlining Background, current situation, problem, benefits, **risks for doing this project** and risks of not doing this project to Victoria detailing the need for an integrated information system.**

- **Project Cost and Delivery Timelines**

The management information system is likely to **cost around 2 million** and a **implementation time of 3 months**.

Risks of doing the project.

- 1. Increased load** on existing school infrastructure
- 2. Resource constraints**
- 3. New system too complex for admin staff to use**
- 4. System does not meet the requirements**
- 5. Database upgrade** to be needed to cope up with future demands

Q1: Write a Business Case outlining Background, current situation, problem, benefits, risks for doing this project and **risks of not doing this project** to Victoria detailing the need for an integrated information system.

Risks of Not Doing the Project.

1. **Errors** arising out of manual process
2. **Duplication of data** being entered.
3. **Inefficiency** in the school administration process
4. **Too much time** taken to copy data from one source to another.
5. Risk of **data being lost** in manual process and data security.

The **project success is to be measured by** the following

- The **decrease in time** spent by the **admin time** due to the automated process.
- **Cost benefits** out of efficiency of process. Automation of data and reduction of manual process.

Q2. Conduct a SWOT analysis on Willow brooks proposal to outsource the administrative process to PWC and co.

Strength	Weakness
PWC have been in the administrative system for years and have a good record of quality and delivery until recently	PWC lack of data security , due diligence and confendiality
Outsourcing the tasks helps the school reduce costs in administrative tasks	Can be a hindrance in part of the schools future development plan
Opportunity	Threat
Outsourcing of the schol administrative tasks is an opportunity for utilisation of resources for other future projects planned by the school.	Leakage of information
Outsourcing also meaning in reduction of cost benefit where saved cost could help in other areas of improvenemt for the school	Competitors schools also outsource with PWC



Willow Brook School – Background

- Based on your earlier recommendations, Willow Brook decided to continue the systems development process for the new information system that is **user friendly, would improve operations, reduce costs and some of the administrative workload. The school has identified that this new system should be operational in 3 months** so that the staff are able to use it and are familiar with it completely before the start of the next session.
- Considering the requirements identified and timeline, it is important to identify the various alternatives available to acquire this new information system. **Options are in-house development, a commercial off the shelf package, customization or Software as a Service.**
- Now, we are in the **systems analysis phase** and before system requirements are captured it is important to **analyze system development strategies** and give a presentation with the recommended option to the Willow Brook School's management and board of directors. Many of the proposed system's advantages were described during the fact-finding process.
- You must examine tangible costs and benefits to determine the **economic feasibility** of multiple alternatives. The finance and fund-raising committees will need to examine the costs and benefits to determine if the development process will go ahead. If Willow Brook School decides to go ahead with development, the system can either be developed in-house or purchased as a vertical package and configured to meet the needs of the school.

Willow Brook School – In House

- Based on your research, you originally estimated that by **working 35 hours per week an in-house development project can be completed in about 8 weeks.** Your **consulting rate** as agreed by Willow Brook, is **\$35 per hour.** One of the options is to design the new system as a database application, which **can be acquired at approximately \$1,500 for a networked commercial package.** One of the parents at Willow Brook has extensive experience with this package as a Database Administrator at a local company, and has committed to voluntarily providing 10 hours per week working with you to develop the project. After the system is operational and the staff is trained, Willow Brook staff and volunteers have agreed to handle all routine maintenance tasks. If they continue to require your assistance, a contract rate can be negotiated later depending on the nature of the problem.

Willow Brook School – Outsource

- As an alternative to in-house development is a SAAS software package called “School-Admin”, **available for about \$8,500**. The software package has been used in schools within the country and is known for providing a quality solution to school administrative needs whilst the support is expensive. The system is proven for years to be quite robust and scalable and is supported by testimonials, indicating an overall tilt towards using their software.
- The vendor offers “School-Admin” as a lease-purchase package of \$2,000 down payment, followed **by two annual installments of \$3,250 each**. If Willow Brook buys the package, it would take you about three weeks of full time work to install, configure, and test. The vendor provides free support during the first year of operation, further-on Willow Brook must sign a technical support agreement at an **annual cost of \$750 for the duration of years**. The package contains not all but many of the features that Willow Brook requires however most of the reports are pre-designed and layouts difficult to modify. The requirements of the integrated information system required by Willow Brook are unique.

Willow Brook School – Outsource (Cont.)

- You have agreed to provide five hours per week of training and support for the first two months of operation. After the new system is operational, it will need routine maintenance, backups, and updating. These tasks will require about **four hours per week** and will be performed by school staff or parent volunteers at an **average hourly rate of \$12.50**.
- For both the in-house development and the SaaS software package, the necessary **hardware will cost about \$3,500**. Hardware and network installation will be done by three parent volunteers.

In order to recommend a development strategy to Willow Brook management, there is need to analyze whether to go for In-house development or SaaS software package. 1. **To conduct this analysis, perform a total cost analysis of both the options and make recommendation as to which option should be chosen.** 2. Justify the reasons behind your recommendation

Total Cost of Ownership Calculator Willow brook

Gray cells are calculated for you. You do not need to enter anything into them.

Cost per year	Inhouse	Outsource	Outsource savings	Notes for Inhouse	Notes for Outsource
Consultancy Costs by you	9,800.00	3,675.00	6,125.00	35 hours *35 per hour for 8 weeks	35 hours * 35 per hour for 3 weeks
Cost of System	1,500.00	8,500.00	-7,000.00	Network application for 1500	Of the shelf package for 8500
Yearly Maintenance	2,600.00	750.00	1,850.00	12.50*4 hours * 52 weeks	Yearly is 750
Hardware	\$3,500.00	\$3,500.00		Same cost	
Total	\$17,400.00	\$16,425.00	\$975.00		

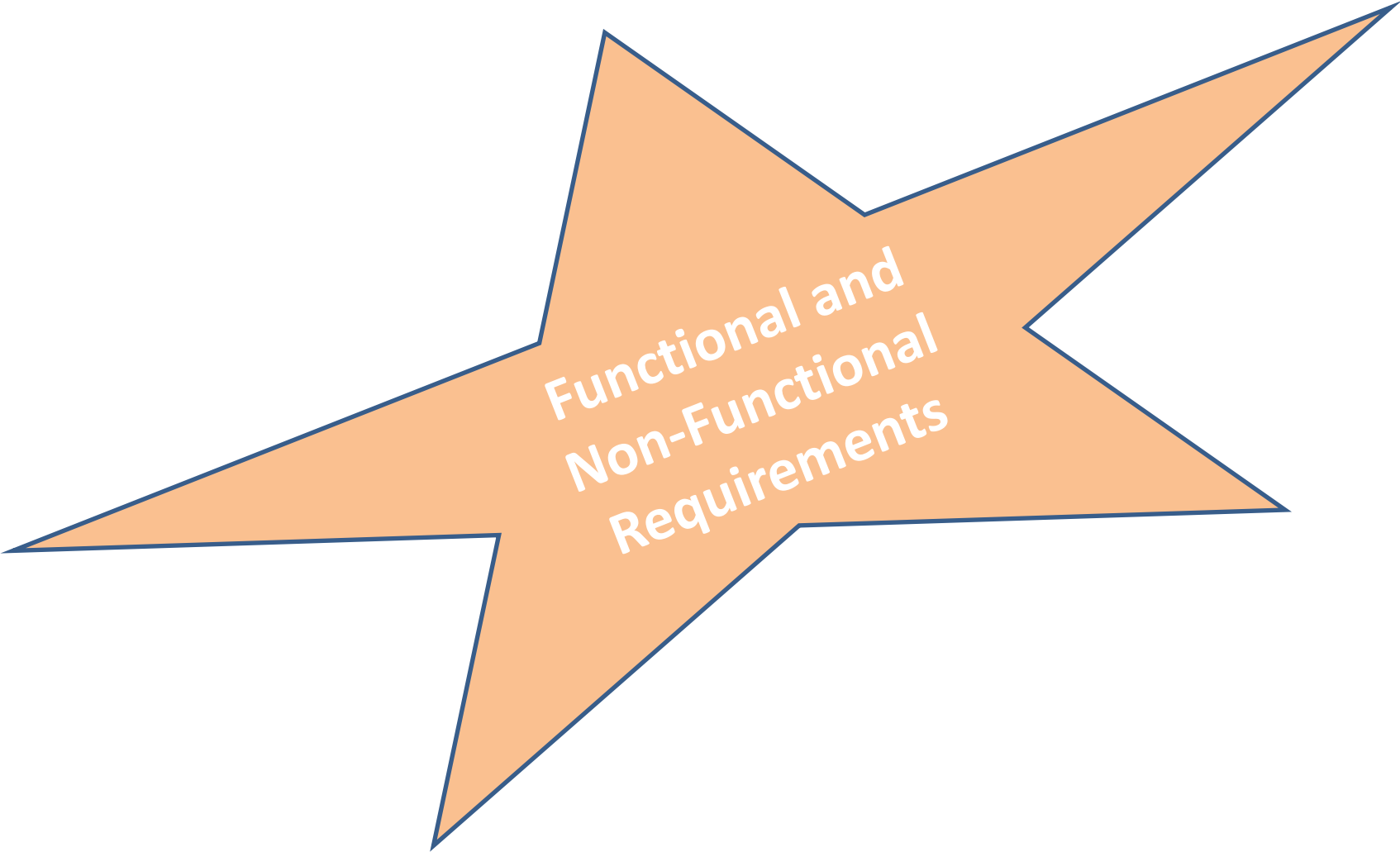
In order to recommend a development strategy to Willow Brook management, there is need to analyze whether to go for In-house development or SaaS software package. 1. To conduct this analysis, perform a total cost analysis of both the options and make recommendation as to which option should be chosen. 2. **Justify the reasons behind your recommendation**

The factors considered were the following

- Time to develop and implement the product
- Skillset of maintenance staff for in house development.
- Indications in the case study which has references to good feedback on the off shelf product.
- Training documentation and system manuals will be readily available for the off the shelf product while for the in house product it has to be created

Considering TCO and other factors as listed above for Willow Brook it is recommended to use the off the shelf product for the following reasons

- Implementation time is faster compared to in house by 5 weeks
- The off the shelf product provides free support for a year
- Yearly Maintenance is cheaper compared to in house
- Also since the off the shelf product has been existing for some, it is highly likely to be quality product with less defects



**Functional and
Non-Functional
Requirements**

Willow brook School – Functional Requirements

- **Registration** for Willowbrook School has two components, **regular daily students and children in after-care**. Regular daily students are divided into groups, dependent on whether the child is in pre-school, kindergarten, or primary school. **Children in after-care are divided into two groups**, those students who are pre-registered and those who use the service on a “drop-in” basis. Most of the students who are pre-registered for after-care attend the school during the day but some are bussed in from other schools. For a student to be registered, tuition payments must be in good standing.
- There are two payment options for tuition and the pre-registered component of after-care, monthly or weekly. **Regular daily tuition** is based on the academic program in which the student is enrolled. A discount of **5 percent** is applied to fees for parents choosing the monthly payment option. Parents who have more than one child enrolled at Willowbrook are eligible for an additional 5 percent multi-child discount on tuition for all enrolled children. Discounts are lost for any payment that is overdue by more than seven days.

Willow brook School – Input & Performance

- Charges for after-care used on a “drop-in” basis are based on an hourly rate, billed in fifteen minute increments. Parents are billed monthly for regular students, and charges are added to the monthly and weekly bills. New bills are generated weekly for “drop-in” care for children who are not pre-registered for after-care or regular daily students. For parents who pay monthly, tuition bills and after-care fees are due on the first of the month, and are distributed to parents one week prior to their being due. For parents who pay weekly, payments are due on Mondays with bills generated on Fridays. Parents who have more than one child enrolled in Willowbrook receive a separate bill for each child. The student’s account must be in good standing for a student to be registered. For the account to be considered to be in good standing, the tuition payment must not be over seven days late. Tuition payments that are over seven days late may be overridden by the director; these cases are handled on a case-by-case basis.

Willow brook School – Functional Requirements

- A summary monthly report is generated for the Board of Directors from this spreadsheet summarizing fee collections for tuition and after care. The financial committee, who provides regular oversight for the Board, also receives separate monthly reports detailing payroll expenses for all employees. Reports need to be ready for distribution at the financial committee meeting that is scheduled for the second Tuesday of every month.

Willow brook School – Process

- Michelle has indicated that the most useful new feature that could be added to the system would be the ability to generate one bill per family, listing each student's charges separately, and an overall total. Also useful would be the ability to generate a receipt at the time payments are recorded. This receipt should contain information required for reimbursement requests from different types of dependent care and tuition accounts, including name, location, and federal tax ID number of the school. In addition, parents have asked for a separate statement accompanying their bill itemizing all drop-in after-care hours for that statement.

Q1: List the system requirements with examples for each category: Output , Input, Process, Performance, Control

- **Outputs:** The system must generate a receipt at the time payments are recorded.
- **Inputs:** The data entry screen for drop-in after care should allow for entry of multiple drop-in after-care uses including after-care start time and checkout time.
- **Processes:** The system should be able to produce a single bill for families with more than one child at Willow brook.
- **Performance:** The system must prepare billing statements for distribution every Thursday for weekly payers and one week prior to the first of the month for parents who pay monthly.
- **Controls:** While not explicitly mentioned, the system needs to have security to protect financial data about the school, data from payment transactions, and protections to safeguard the identities of the children.

Q2: Identify the requirements from the given facts in the case study here and classify them as **Functional** and **Non-functional**.

- System must handle registration for regular students and children in after-care.
- Must track the payment method and status of students including calculation of payments and amount due.
- System must calculate discounts.
- Should have an option to allow the director to override accounts that have overdue payments.
- Must flag accounts that are overdue.
- Minimum data input into the system should include data fields currently input into the Excel spreadsheets.
- The system should provide itemized bills that break down fees by type, together with a grand total for the month.
- Generate a summary report for the finance committee for the second Tuesday of the month.

Q2: Identify the requirements from the given facts in the case study here and classify them as Functional and **Non-functional**.

- Non-functional requirements cover all the remaining requirements which are not covered by the functional requirements. They specify criteria that judge the operation of a system, rather than specific behaviours, for example: “Modified data in a database should be updated for all users accessing it within 2 seconds.”
- Some typical non-functional requirements are:
 - Performance – for example Response Time, Throughput, Utilization, Static Volumetric
 - Scalability, Capacity, Availability, Reliability, Recoverability, Maintainability, Serviceability, Security, Regulatory, Manageability, Environmental, Data Integrity, Usability, Interoperability

Q3: Are there scalability issues that you should consider? If so, what are they?

- Scalability refers to a system's ability to handle increased business volume and transactions in the future. To quantifiably evaluate scalability, you will need information about projected future volume for all outputs, inputs, and processes. Scalability considerations for Willowbrook should include adding new functions to the system including the ability to handle increasing enrollment, add summer camps, payroll, and student records to the information system. A full discussion of scalability can be found on page 145.

Q5: Can you suggest more **non-functional requirements** based on your knowledge of the Case study?

- From the above case study, most of the requirements are from a functional perspective.
The only non functional requirement would be the security requirements in protecting data of the school.
Other non functional requirements possible are
Performance – The billing statements should be generated in no greater than 15 minutes from the system
- **Scalability** – The schools information system should be capable of handling student intakes up to 15000 students per annum.
- **Security** – Ensuring that the system conform to the ISO 27001 standards of data security and prevention
- **Regulatory** – All reports and bills generated by the system conform to the Australian regulatory requirements.



User Story and Use Cases

Willow brook School

- **In agile environments, less of the design is done upfront and more of it is delegated to component teams.** The key is determining how deep to go initially, which decisions to defer and identifying when decisions need to be made. Decisions that impact multiple development teams should be made earlier, **especially scalability and security.** For example Decisions like adding additional languages to an already internationalized product can be deferred until very late.
- After the initial design is created, the architect works with each of the teams and reviews their designs. If additional design or design changes are required for a unit of work (such as a scrum sprint), the architect aims to have it available by the time that unit of work starts. The architect is also responsible for communicating any changes to affected teams or stakeholders.

Willow brook School

Case Study Willow Brook School

- Willowbrook School is a small private school that has retained your services as a systems analyst to assist in the development of a new information system for the school's administrative needs.

Background

- Willowbrook School has decided to precede to the systems analysis phase, based on the findings and proposal you presented after the preliminary investigation.
- Michelle the administrative assistant needs a UI to enter the student details where she can add information and edit them if needed and also generate a report on priority which contains the following
 1. First Name
 2. Last Name
 3. Address
 4. Phone

Q1: Write user story 'add new student' along with the acceptance criteria Solution. List the acceptance criteria for the user story.

- **User Story**

As a Parent/Admin staff, I want to add a new student to Willowbrook School

- **Acceptance Criteria**

1. Registration is available
2. Administrator notifies parent
3. Registration is open and parent pays initial tuition/drop in fee
4. Administrator registers student by entering student details

Q2: It is identified that more details are required to design the user story add new student to address the situations where student cannot be registered. Hence it is decided that full use case description need to be created for the user story 'add new student' as on page 236 of the textbook.

ADD NEW STUDENT Use Case	
Name:	Add New Student
Actor/Role:	Administrator
Description:	Describes the process used to add a student to <u>Willowbrook School</u>
Successful completion:	1. Administrator checks <u>REGISTRATION availability</u> 2. Administrator notifies parent 3. Registration is <u>open</u> and parent pays initial tuition/drop in fee 4. Administrator registers student entering student details
Alternative:	1. Administrator checks <u>REGISTRATION availability</u> 2. Class registration is full 3. Administrator notifies parent 4. Administrator offers to add student to wait list
Precondition:	Parent requests child to be enrolled in school or "drop in"
Post condition:	Student is enrolled in academic or "drop in" program and initial fees have been paid
Assumptions:	New parent/student with no relationship to school

Another Example: Use Case (Create Customer Account)

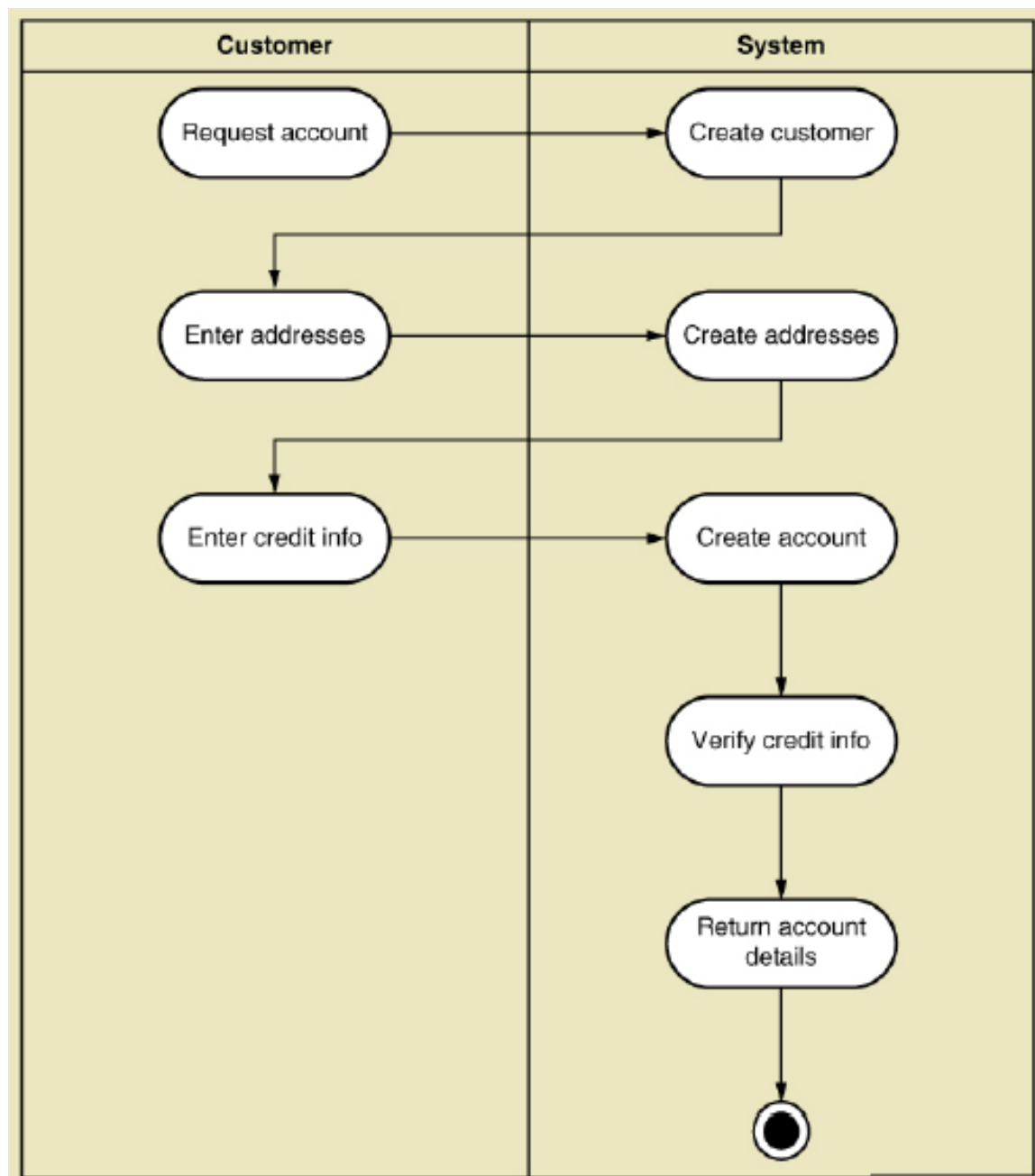
Use case name:	Create customer account.	
Scenario:	Create online customer account.	
Triggering event:	New customer wants to set up account online.	
Brief description:	Online customer creates customer account by entering basic information and then following up with one or more addresses and a credit or debit card.	
Actors:	Customer.	
Related use cases:	Might be invoked by the <i>Check out shopping cart</i> use case.	
Stakeholders:	Accounting, Marketing, Sales.	
Preconditions:	Customer Account subsystem must be available. Credit/debit authorization services must be available.	
Postconditions:	Customer must be created and saved. One or more Addresses must be created and saved. Credit/debit card information must be validated. Account must be created and saved. Address and Account must be associated with Customer.	
Flow of activities:	Actor	System
	1. Customer indicates desire to create customer account and enters basic customer information. 2. Customer enters one or more addresses. 3. Customer enters credit/debit card information.	1.1 System creates a new customer. 1.2 System prompts for customer addresses. 2.1 System creates addresses. 2.2 System prompts for credit/debit card. 3.1 System creates account. 3.2 System verifies authorization for credit/debit card. 3.3 System associates customer, address, and account. 3.4 System returns valid customer account details.
Exception conditions:	1.1 Basic customer data are Incomplete. 2.1 The address isn't valid. 3.2 Credit/debit information isn't valid.	

Another Example: Use Case (Create Customer Account)

Flow of activities:	Actor	System
	1. Customer indicates desire to create customer account and enters basic customer information.	1.1 System creates a new customer. 1.2 System prompts for customer addresses.
	2. Customer enters one or more addresses.	2.1 System creates addresses. 2.2 System prompts for credit/debit card.
	3. Customer enters credit/debit card information.	3.1 System creates account. 3.2 System verifies authorization for credit/debit card. 3.3 System associates customer, address, and account. 3.4 System returns valid customer account details.

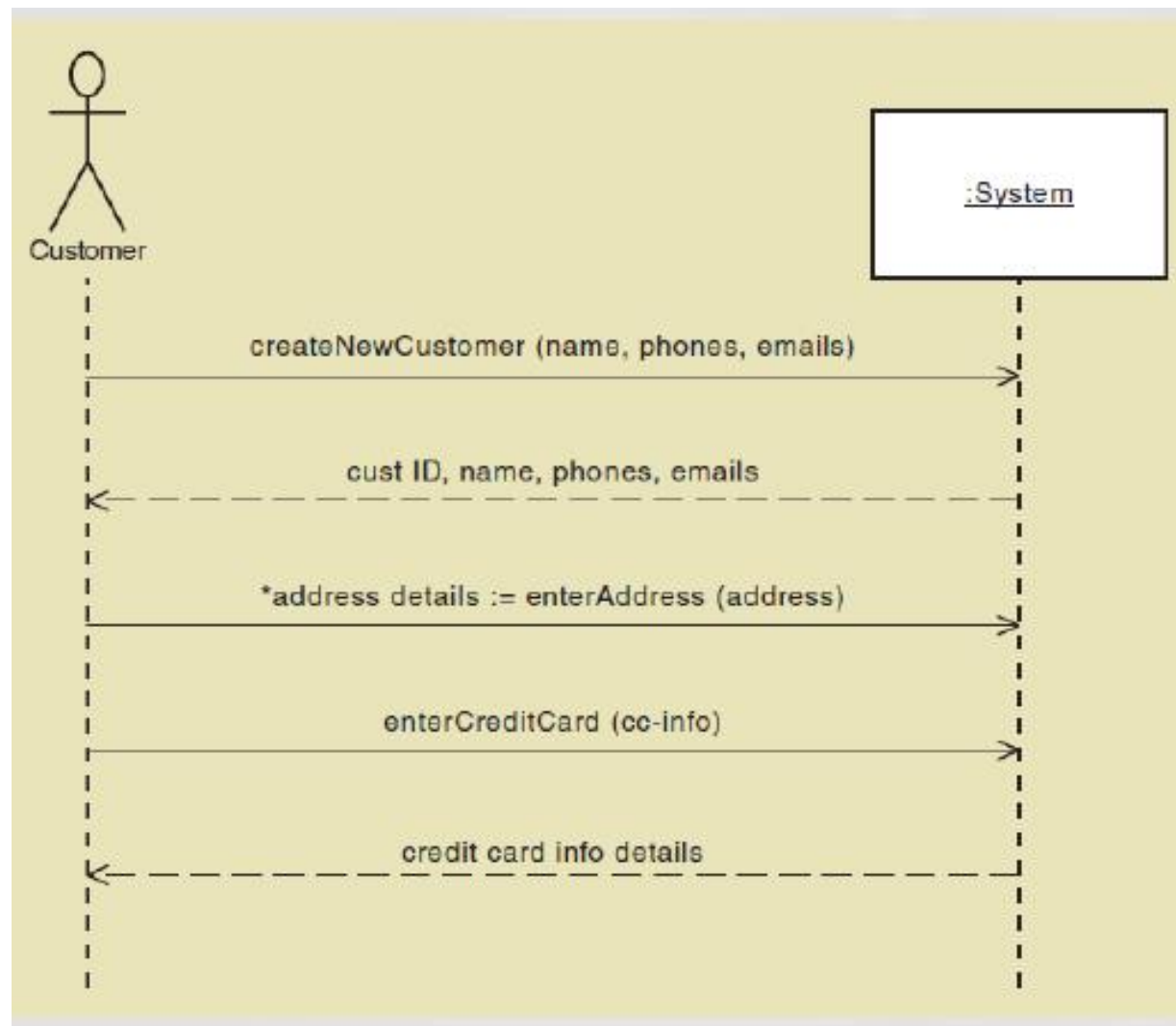
Another Activity Diagram (Create Account)

Example: Diagram Customer



Another Sequence (Create Account)

Example: Diagram Customer

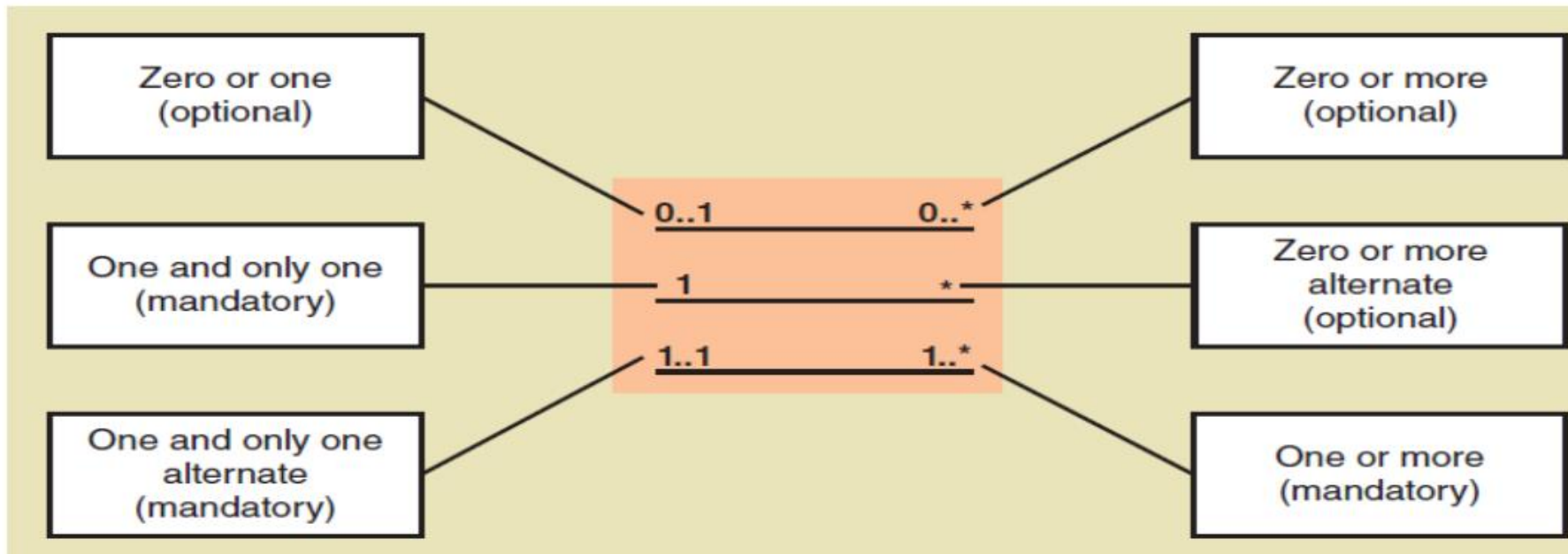


**Entity Relationship
Diagram (ERD) &
Data Flow Diagram
(DFD)**

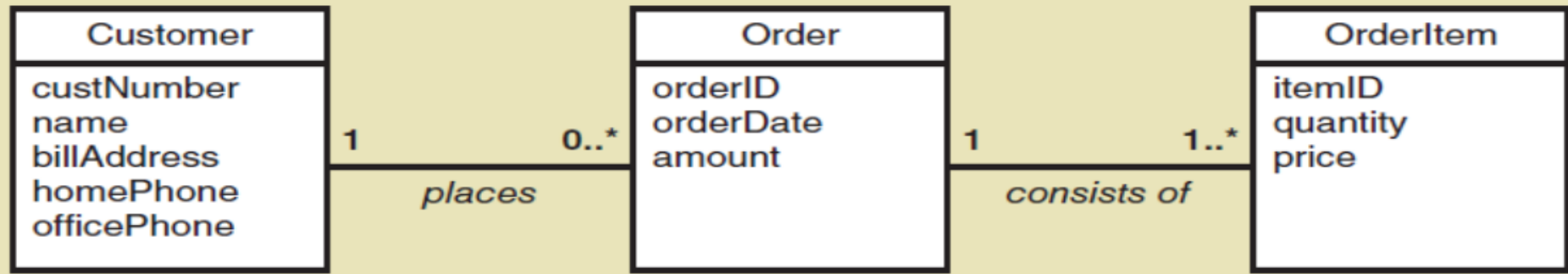
Willow brook School

- Willowbrook School is a small private school that has retained your services as a systems analyst to assist in the development of a new information system for the school's administrative needs.
- **Background**
- Now that you have designed a user interface, you are ready to focus on the data design of the Willow brook School's information system. To perform the following tasks, refer back to the documentation given in earlier weeks and also ones you prepared from the systems analysis phase.

UML Notation for Multiplicity

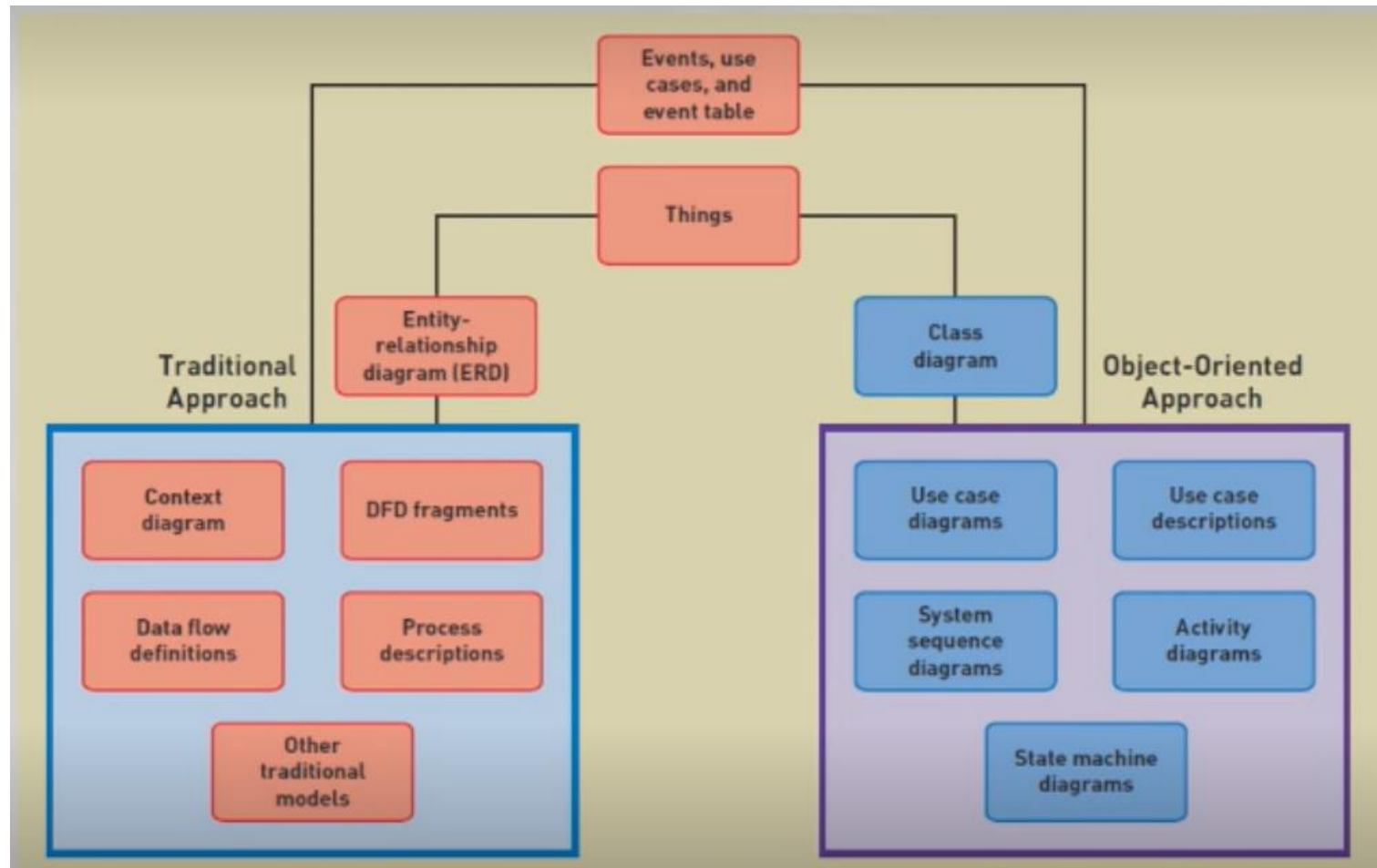


A Simple Domain Model Class Diagram



- Note: This diagram matches the semantic net shown previously
 - A customer places zero or more orders
 - An order is placed by exactly one customer
 - An order consists of one or more order items
 - An order item is part of exactly one order

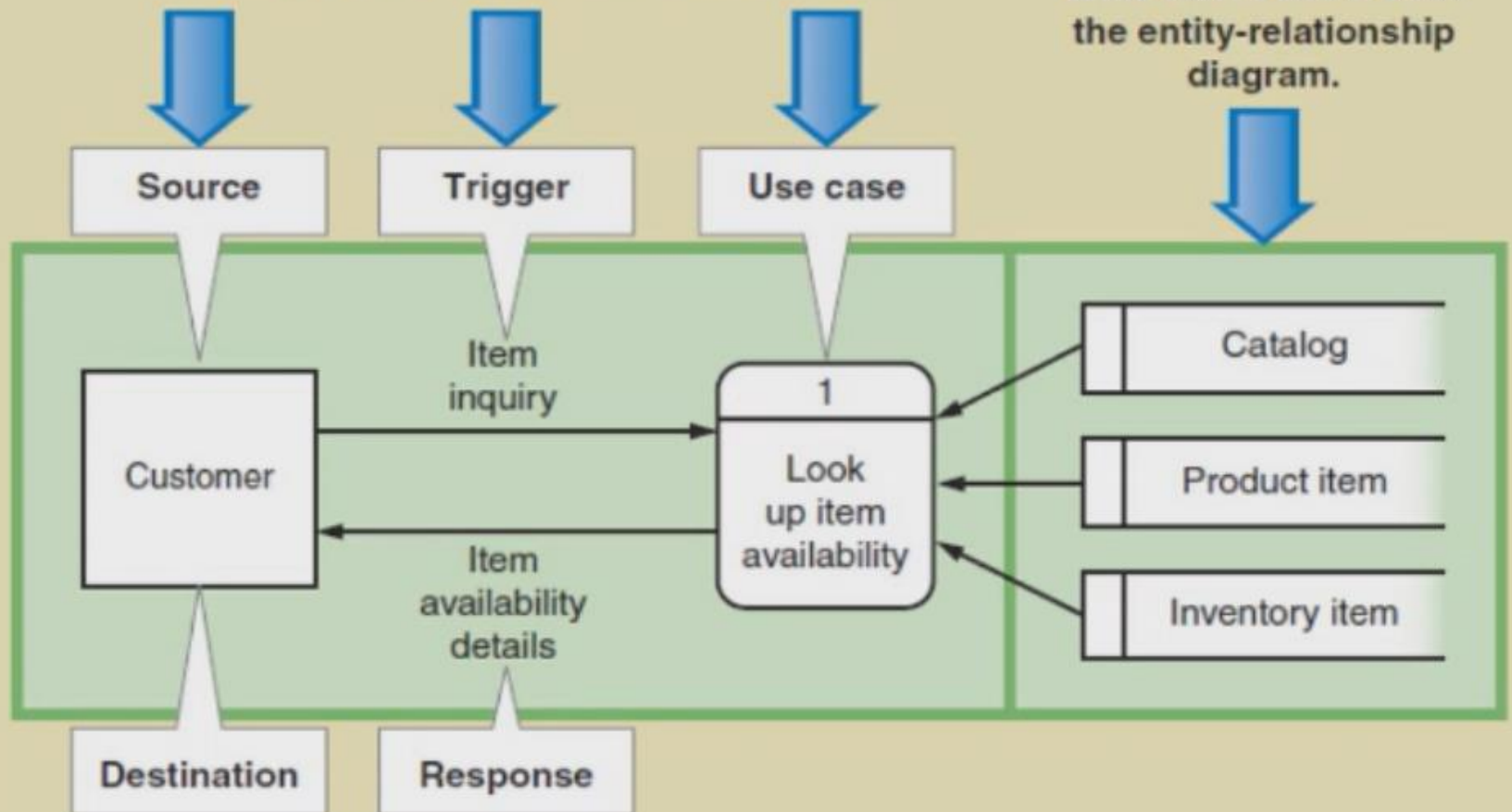
Relationships Between the Traditional and OO Approach



DFD for process *Look up item availability*

External agent, data flows, and the process come from information about the event in the event table.

Data stores come from the entity-relationship diagram.



DFD Decomposes

To show different levels of abstraction

Context Diagram
Diagram 0
Diagram 1

Context Diagram

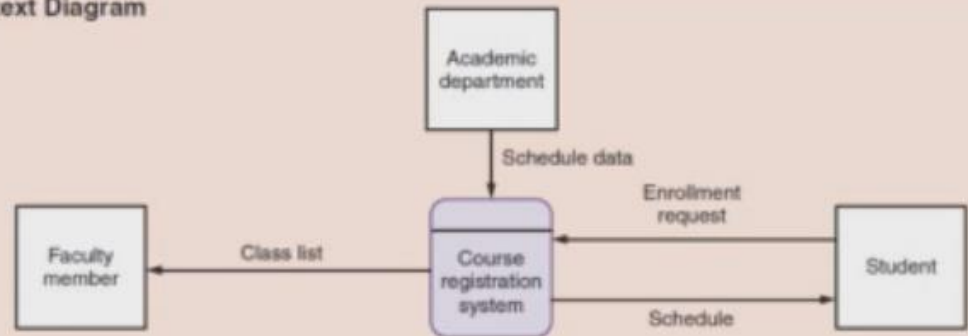


Diagram 0

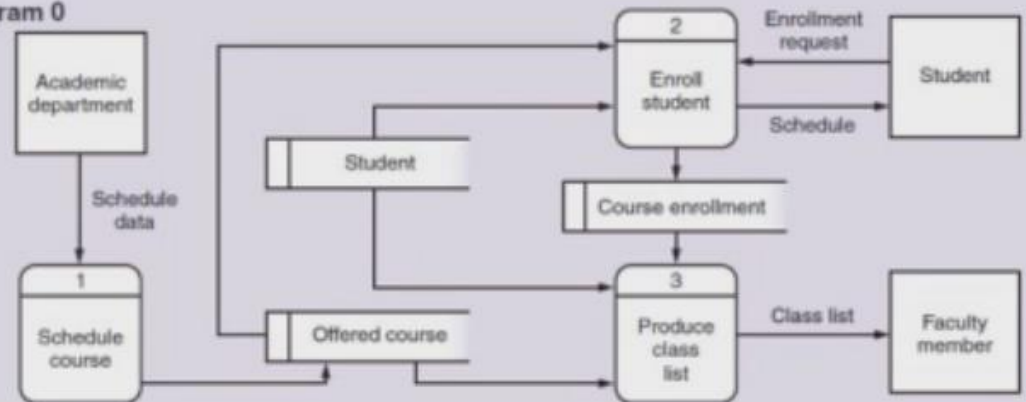
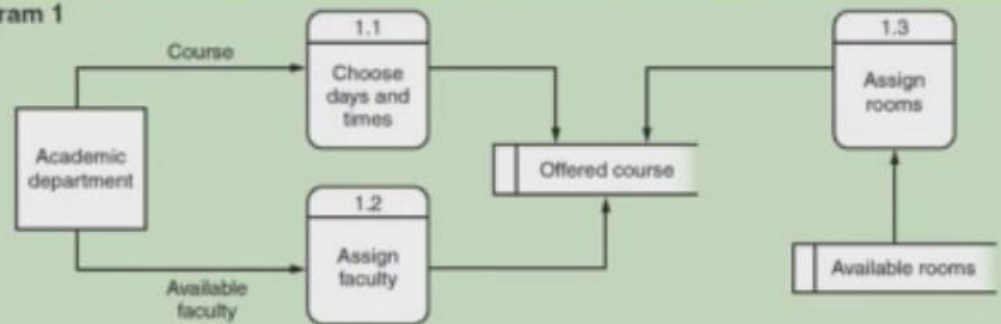
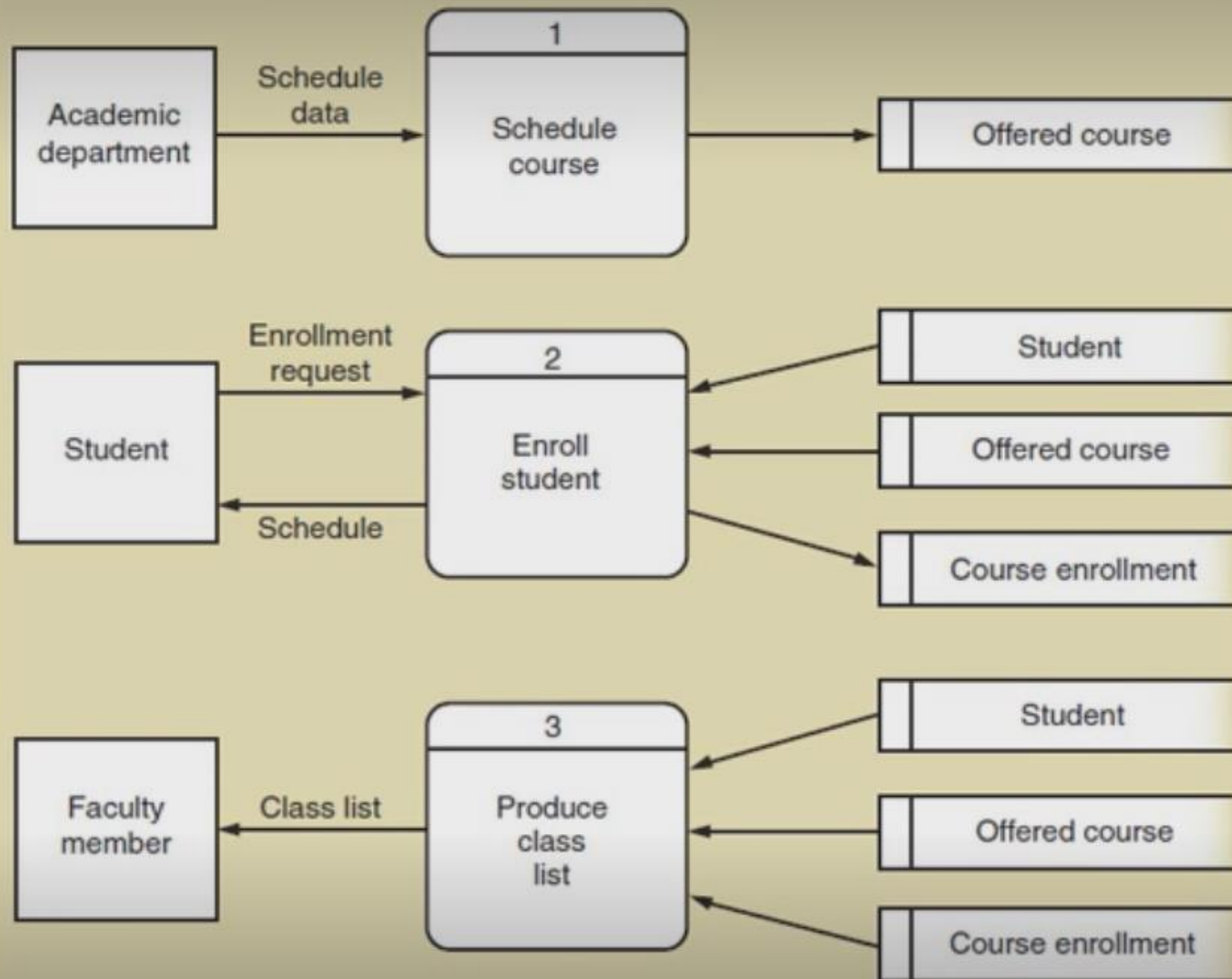


Diagram 1



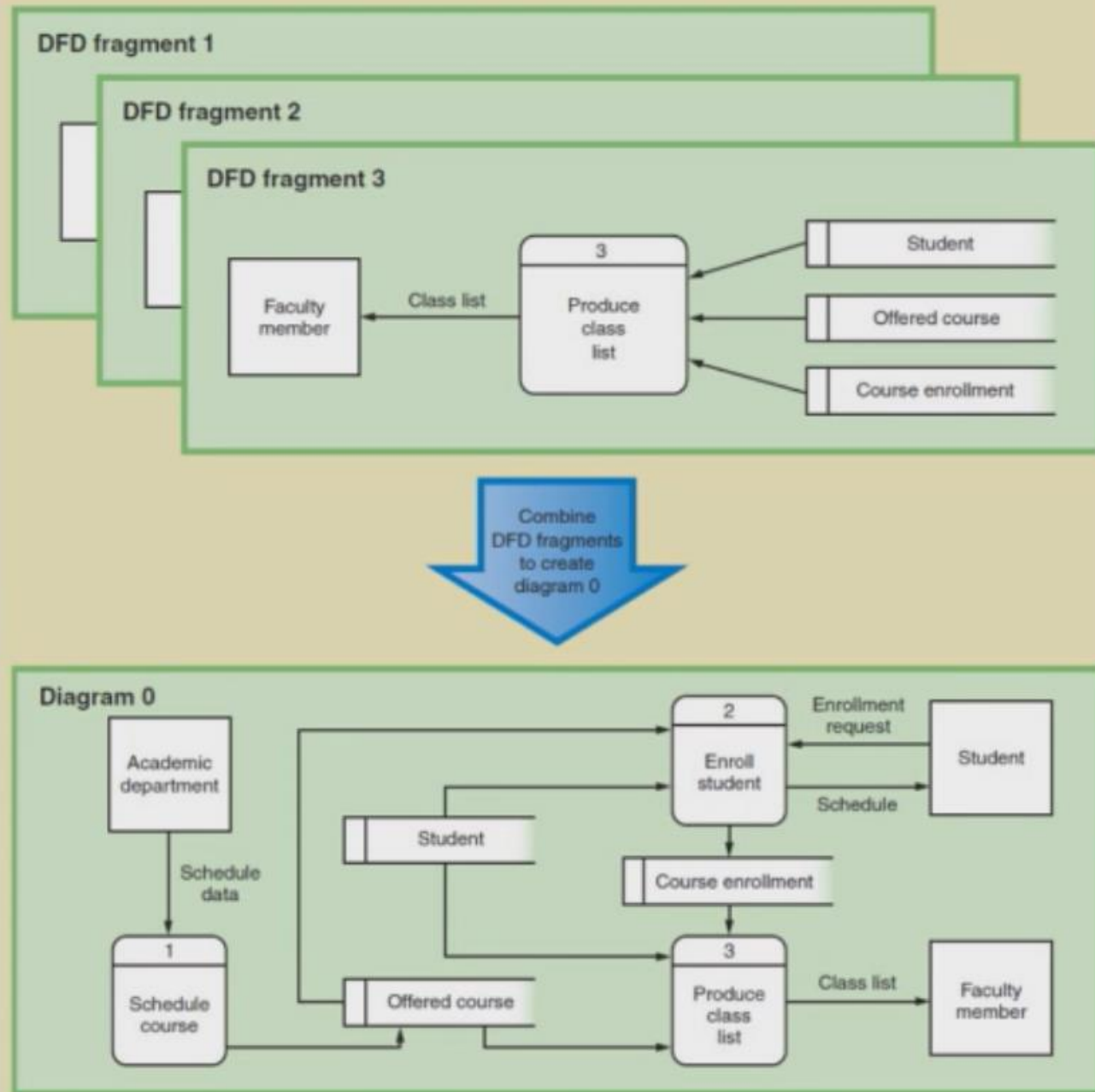
DFD Fragments

One for each
use case

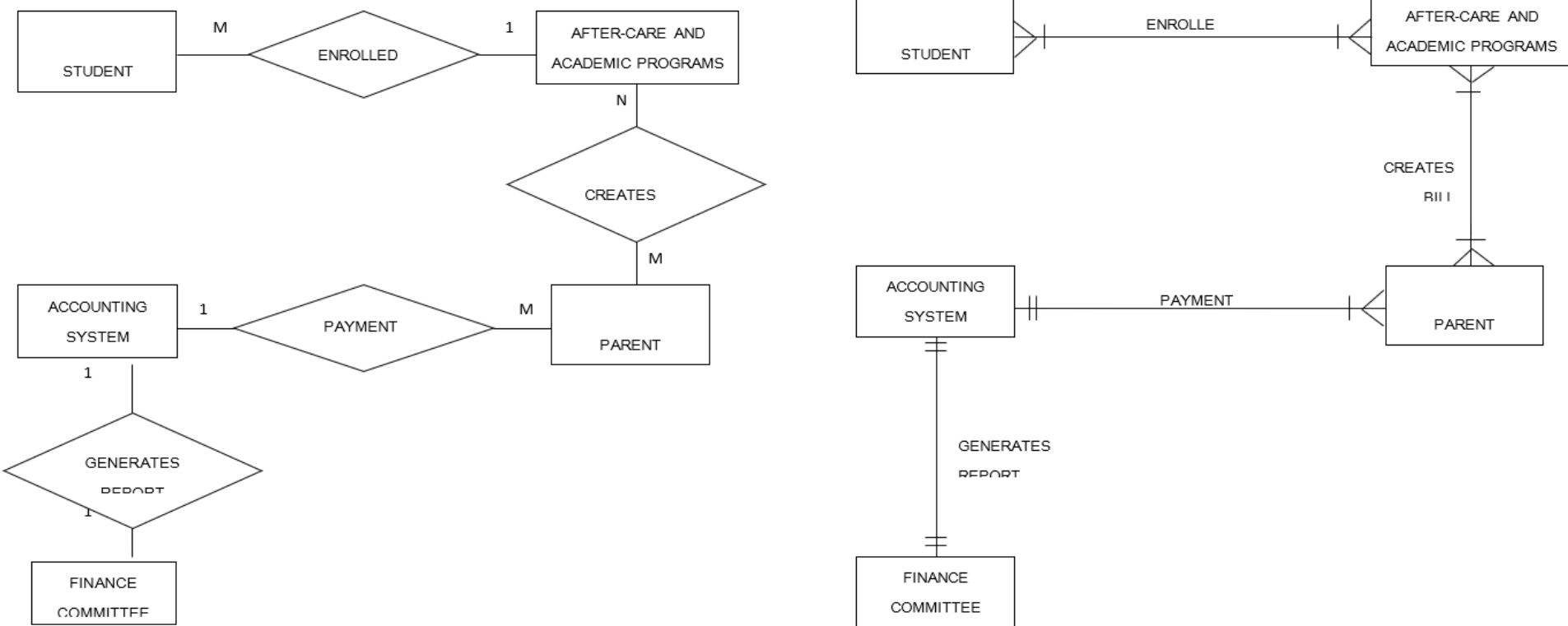


Combine DFD Fragments

To Create Diagram 0



Q1: Create an ERD for the Willow brook School's information system showing relationships and cardinalities and notation as explained in Data Design lecture recording



Q2: Design fully normalized 3NF tables for the system.

Unnormalized

*Student (Student#, Student_Name, Monthly_Fee, (Household#, Parent_FIRST_NAME, Parent_LAST_NAME, ADDRESS, CITY, STATE, ZIP, PHONE1, PHONE2)
(Date_Of_Attendance, Classroom_Attended))*

1NF

The key for the student table should be expanded to include the repeating rows of the date of attendance.

Student (Student#, Household#, Student_Name, Monthly_Fixed_Fee, Parent_FIRST_NAME, Parent_LAST_NAME, ADDRESS, CITY, STATE, ZIP, PHONE1, PHONE2, Date_Of_Attendance, Classroom_Attended)

Q2: Design fully normalized 3NF tables for the system.

2NF

To be 2NF, a new table should be created each of the fields that are entirely dependent on a key field.

Student (Student#, Household#, Student_Name, Monthly_Fixed_Fee)

pPROGRAM (DATE_OF_ATTENDANCE, STUDENT#, Classroom_ATTENDED)

Parent (Household#, FIRST_NAME, LAST_NAME, ADDRESS, CITY, STATE, ZIP, PHONE1, PHONE2)

Q2: Design fully normalized 3NF tables for the system.

3NF

Since a student may attend an academic class and an after-school class on the same day, there should be an additional classroom table.

Student (Student#, Household#, Student_Name, Monthly_Fixed_Fee)

pPROGRAM (DATE_OF_ATTENDANCE, STUDENT#, ClassRoom #, Program_NAME)

Classroom (Classroom#, TEACHER_NAME)

Parent (Household#, FIRSTNAME, LAST_NAME, ADDRESS, CITY, STATE, ZIP, PHONE1, PHONE2)

Q2: Design *fully normalized 3NF tables* for the system.

Sample data for 3 STUDENT records might look like this:

Student#	Household#	Student Name	Monthly FixedFee
101-1	101	Owen, Susan	\$375
101-2	101	Owen, Michael	\$350
104-1	104	Cohen, Emma	\$375

Sample data for 3 PROGRAM records might look like this:

DATE OF ATTENDANCE	Student#	CLASSROOM#	PROGRAM NAME
13NOV14	101-1	2201	First Grade
13NOV14	101-2	2203	Third Grade
13NOV14	104-1	4120	After-Care

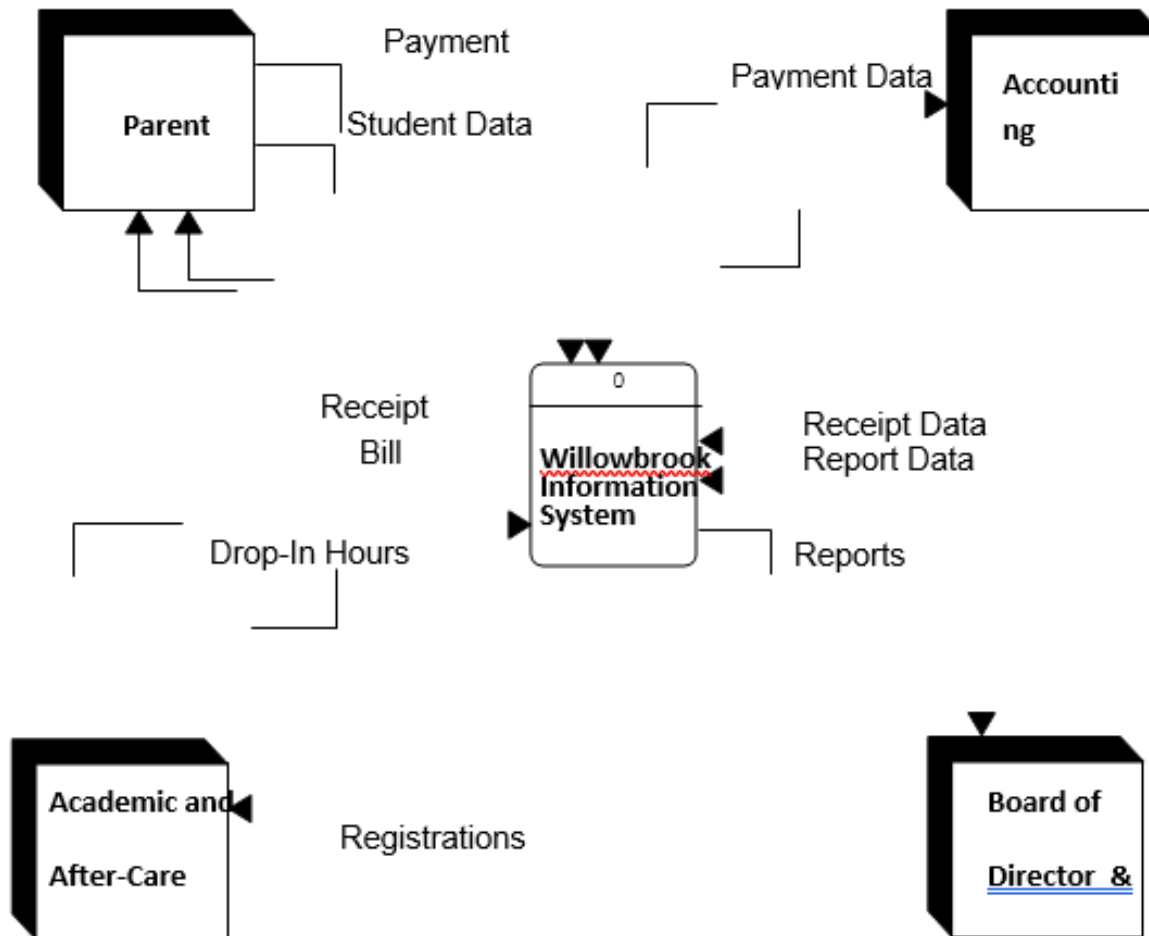
Sample data for 3 CLASSROOM records might look like this:

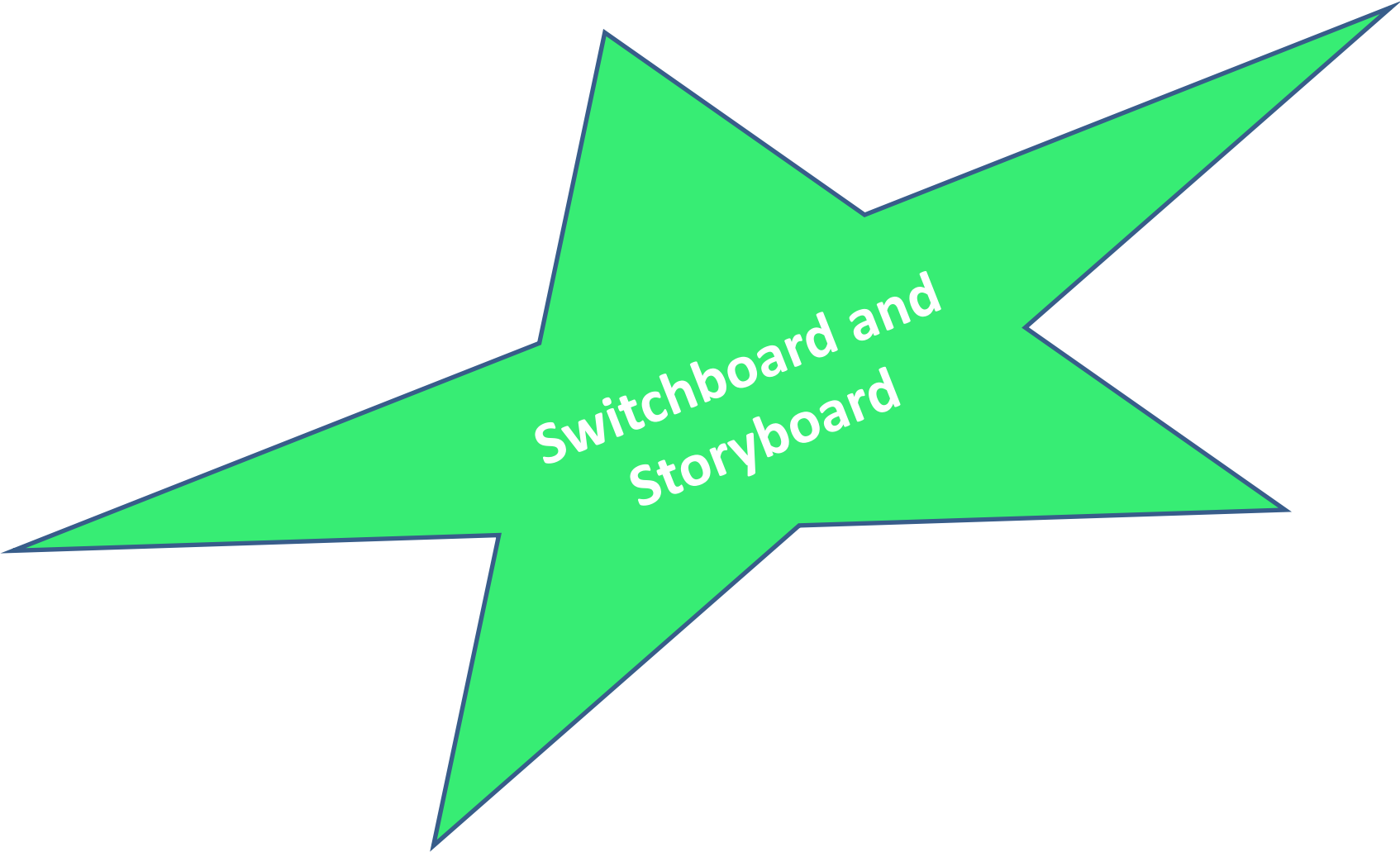
Classroom#	TEACHER_NAME
2201	Foxwood, Susan
2203	Shapiro, Jason
4120	Bell, Eguanna

Sample data for 3 PARENT records might look like this:

Household#	FIRSTNAME	LASTNAME	ADDRESS	CITY	STATE	ZIP	PHONE1	PHONE2
101	Mary	Owen	1 Main St	Ourtown	IL	60572	5551111	5552222
103	Rudy	Forsythe	12 Elm St.	Mytown	IL	60290	5555555	5556666
104	Mark	Cohen	4 Eve St.	Ourtown	IL	60572	5553333	5554444

Q4: Draw a context diagram for willow brook Information system (IS)





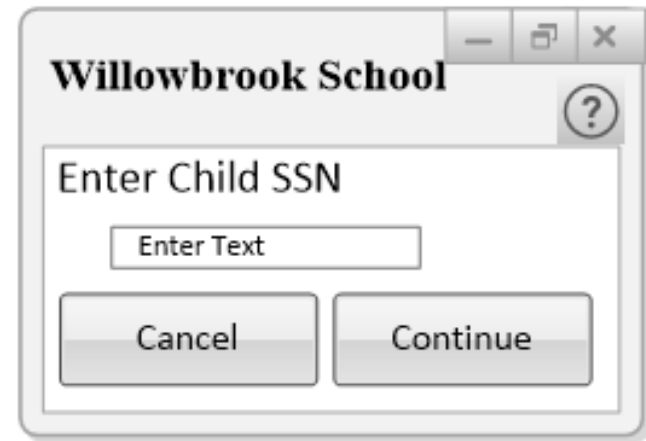
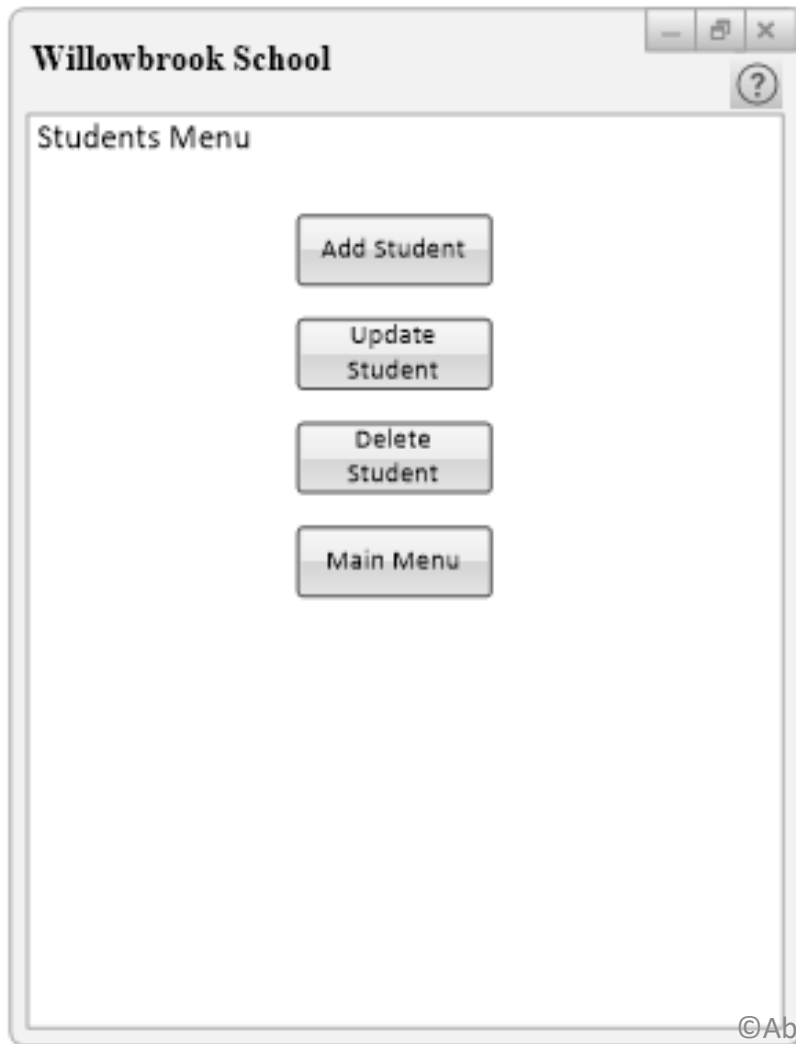
Switchboard and Storyboard

Willow brook School

Background

- After presenting the system requirements document to Willowbrook, the staff has approved your recommendations for a new computer system. Last workshop we began the design phase of the SDLC and designed UIs for Michelle. This system for school would contain components such as students, staff, classrooms, after-care, payments and any other area that may be recognized in sprint review to be included in a working system.
- This workshop we would design user interface including screens and forms for inputs, navigation and reports for outputs.
- Please review the documentation you prepared during the systems analysis phase such as mind maps, user stories, and process diagrams and design these user interfaces for Willow brook.

Q1: Create a **switchboard design with appropriate control buttons that lead to system component namely student. Please make sure that you consider any other area you think should be in a working system of your own design related to workflow related to UI for Michelle to enter & edit student details and generate reports.**



Q2: Prepare **storyboards** that show lower-level menus including data entry screens. Allow Michelle to add, update, or delete student. Suggest context-sensitive and specific help and feedback for the proposed screens.

Willowbrook School

Add Student

Contact Info

First Name: Enter Text Middle Name: Enter Text
Last Name: Enter Text DOB: Enter Text

Emergency

Parent / Guardian information:

First Name: Enter Text Middle Name: Enter Text
Last Name: Enter Text DOB: Enter Text

Dietary

Address line 1: Enter Text
Address line 2: Enter Text

Education

City: Enter Text State: QZ Zip: Enter Text
Phone 1: Enter Text Phone 2: Enter Text

Main Menu

First Name: Enter Text Middle Name: Enter Text
Last Name: Enter Text DOB: Enter Text

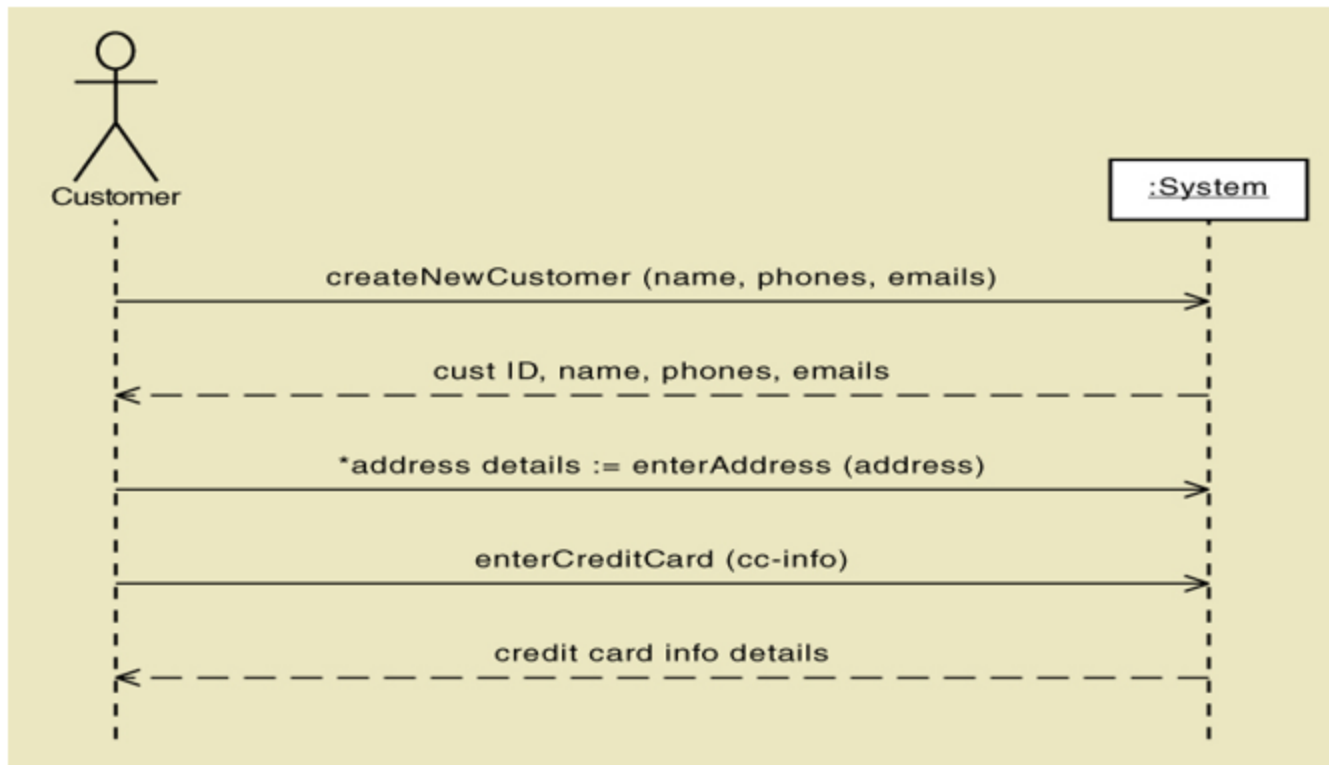
Address line 1: Enter Text ☒ Same as above?
Address line 2: Enter Text

City: Enter Text State: QZ Zip: Enter Text
Phone 1: Enter Text Phone 2: Enter Text

©Abeer Alsadoon

Q3: Give a sample of **context-sensitive help** for the Willowbrook system

<i>Area</i>	<i>Specific Task</i>	<i>Suggested Help</i>
<i>Students</i>	<i>Add Student</i>	<i>Remind user that the SSN is a primary <u>key</u>, and will not allow duplicates. If the student is already in the system, the message "This Student Already Entered" will be displayed.</i>
	<i>Update Student Information</i>	<i>Inform user about various data validation checks that apply to fields in this area.</i>
	<i>Delete Student</i>	<i>Inform user that, without specific confirmation, the system will not allow a student to be deleted if that student currently is registered in one or more programs or activities. The user first must click the button labeled "OK to delete student and cancel activities."</i>



Customer Form



Customer ID

Customer Name

Mobile Phone

Home Phone

Email Address

Save

Cancel



Use case *Check out shopping cart*

SYSTEM: What would you like to do?

USER: I'd like to check out.

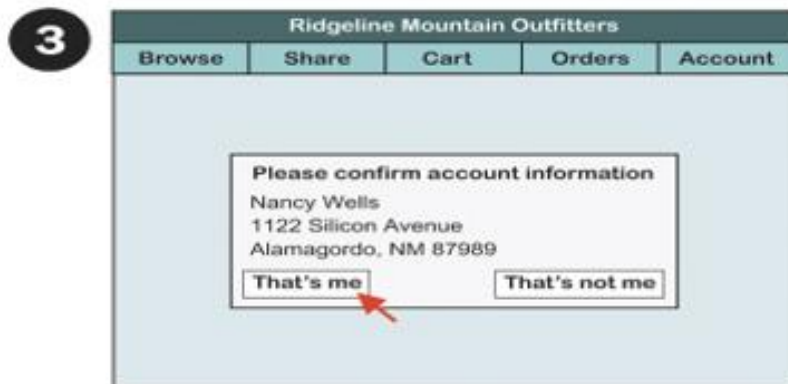
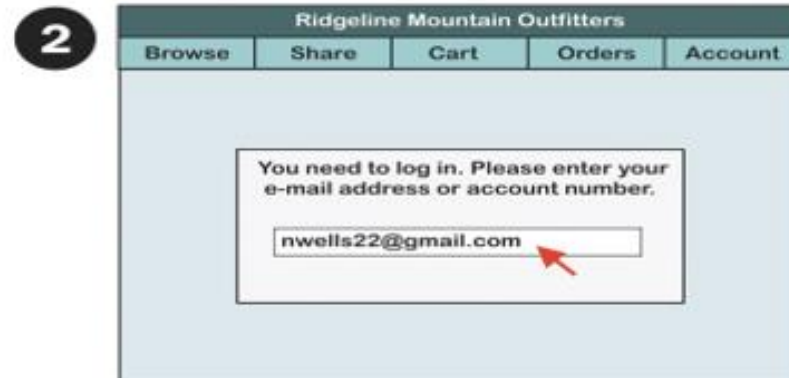
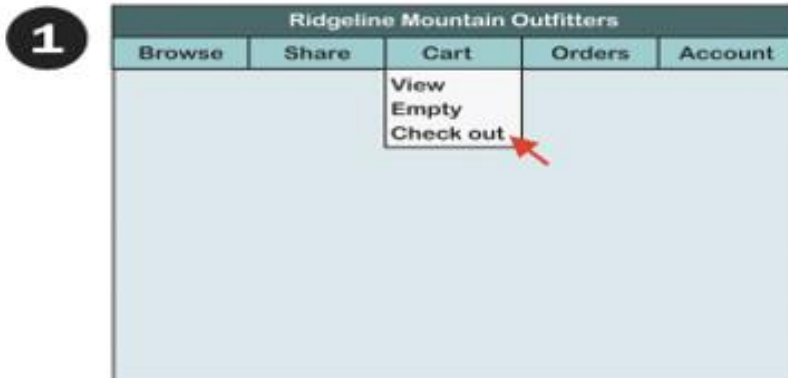
SYSTEM: Okay. What is your e-mail address or account number?

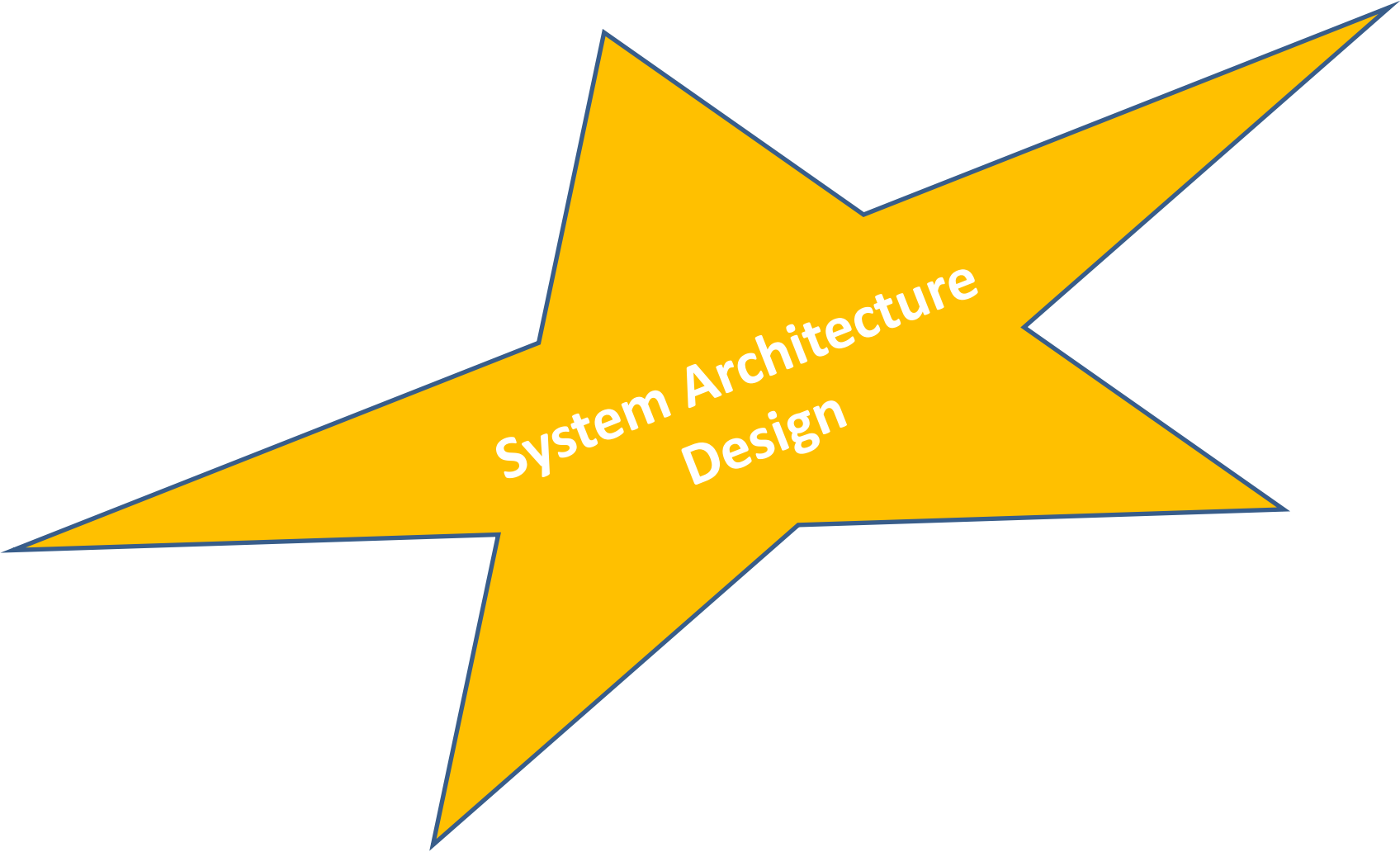
USER: My e-mail address is nwells22@gmail.com.

SYSTEM: Fine. You are Nancy Wells at 1122 Silicon Avenue. Correct?

USER: Yes.

SYSTEM: All items in your cart are in stock and can be shipped today. Because your order subtotal is over \$100, you qualify for free UPS ground shipping (3–5 days). Other shipping options include next day (\$35.00), two day (\$20.00), and USPS parcel post (\$11.70, 5–7 days). How would you like your items to be shipped?





System Architecture Design

Willow brook School

- **Background**
- The Willow brook School staff accepted your interface, output, input, and data designs. Now you will determine the system architecture, which translates the logical design of an information system into a physical blueprint.

•

The following are the System Requirements from an architectural perspective.

1. **Availability:** The school IS would offer different services to store data related to enrolment requests, admissions and various students and staff activities once students are enrolled in programs at school. **The schools number one Priority is the availability of the data.** Consider that the school stores huge amount of student information and information pertaining to staff and school accounts, it is **of huge importance that data is accessible via the database even if application interface of IS is down** so that school activities can be performed as normal.

Willow brook School

- **2. Data Integrity:** Data integrity requirement varies within the system. The **most important data is student information followed by data related to school such as staff details and school data entries such as attendance details, student activities and library activities.** It is essential that this **data is never lost** and the system must guarantee that **each data is stored correctly and accurately.** In contrast, data lost from the accounting system is easily recovered by information from the bank where there is a record of the transactions made by the school.
- **3. Performance:** The information system is expected to be capable of coping with the increase in numbers as the school grows. The school though does not expect very high levels of performance in comparison to large organizations such as banks.

Willow brook School

- 4. **Security:** The school IS contains personal and financial information about students and staff and hence should be kept secure. This should be done by preventing any unauthorized access to the information system, ensuring secure data transfer between various parts of the system and also by keeping safe administrative policies.
- 5. **Growth:** The school expects to grow in student numbers on an exponential basis and will be doing timely upgrades to its existing infrastructure and database to cater to future growth.
- 6. **Backup and Recovery:** The school IS will contain various types of information about students and staff and the school also understands that the system will encounter problems from time to time. Hence they need mechanisms for recovery from such problems and issues. A key factor in determining the system's success is how quickly the system can recover from failure. Backup and recovery must be smooth and non-disruptive. One way to ensure that the system can easily recover from a system crash is to make sure the data is duplicated elsewhere on the system.

Q1: Design system architecture for Willow brook School based on file server or client/server architecture? Discuss advantages and disadvantages that should be considered when making this decision within the context of Willow brook School.

- Based on all the facts presented thus far, it seems clear that the new system should be based on client/server architecture

Characteristics of Client Server

1. Client-server is a computing architecture which separates a client from a server
2. It is almost always implemented over a computer network
3. The most basic type of client-server architecture employs only two types of nodes: clients and servers.
4. This type of architecture is sometimes referred to as two-tier.
5. It allows devices to share files and resources.
6. Server provides the service
7. Client is considered as the customer requesting the service
8. The server service can be shared among a number of clients
9. Clients must request or initiate the service
10. The location of the server in the network is transparent to clients
11. Transaction between C/S is message-passing based
12. C/S architecture is scalable
13. horizontally (more clients can added)
14. Vertically (more servers can be added)

Q2: From the above requirements, identify the type(s) of testing which could be performed on the Information system?

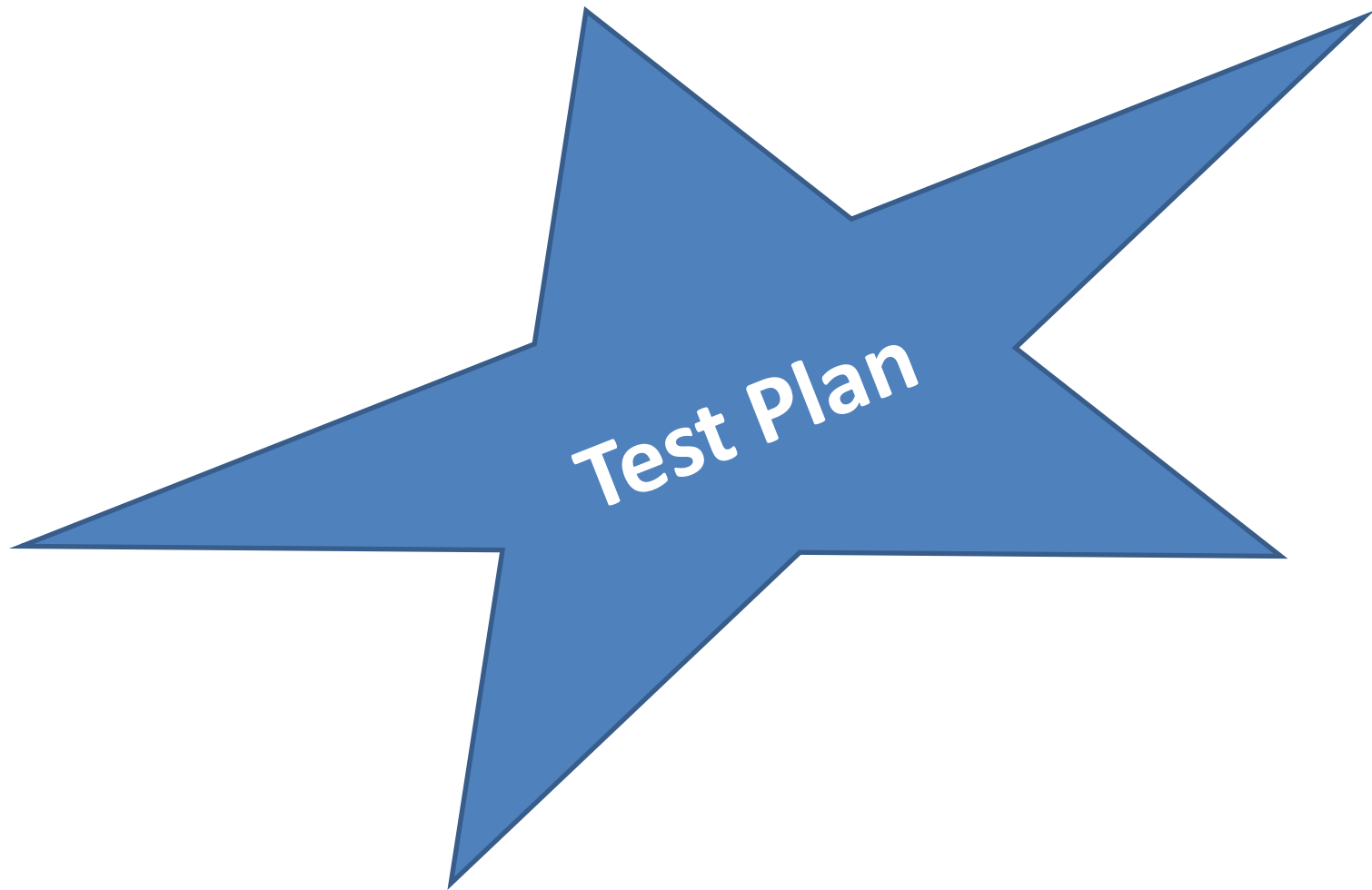
- An effective test plan must include unit, integration, and system testing.
 - A unit test is a test of an individual module or method before it is integrated with other modules or methods.
 - An integration test is a test of the behavior of a group of modules or methods.
 - A system test is a test of the behavior of an entire system or independent subsystem. System testing is performed in a test environment on test computers. The total system, including the database, is compiled and linked together. System testing requires that the system be tested with initial transactions that create and initialize data, update transactions that modify and change the data, with delete transactions that remove or deactivate data. For Willow brook it is largely system testing of the different functionalities of Student registration, Fees calculation, Discount calculation, manager overrides and generation of monthly bills.

Q3: From the case study, identify which data items need to be considered for data conversion? Remember that all data items from the old system need not be converted across to the new system. For example the Serial Number .The new system will by default be assigning a serial number for each student record.

- **For the conversion process**
- The Fees records are the most time critical, because they contain the billing information that must go out at the beginning of the academic year and also contain the information of fees being paid by the students or over dues. The other critical data is the student information which also needs to be moved to the new system. Serial Number field can be left behind as the new system would by default increment the number for each student record. The other data which could be left behind is the total monthly fee which can be calculated by the new system.

Q4: Using your knowledge of System Changeover methods which method is most suitable for Willow brook IS?

- **System Changeover**
- For the system changeover, given the small scale of this operation, and the fact that accounts zero out and start over again in July (save for one or two late payers, who are dealt with on a case-by-case basis as indicated earlier), a direct cutover transition is feasible. Students should note they must include careful checking and double-checking of the first set of bills before they are delivered to the parents. Students may also suggest a Parallel changeover citing the low risk advantages.



Test Cases Writing Steps – “Login”



User Name Administrator

Password *****

Login

[Forgot Password?](#) [Registration](#)

Step 1: get a screen prototype (or wire-frames)

Test Cases Writing Steps – “Login”

- The functional requirements specifications document/s.
 - Either it may be a user requirement document or a functional requirements specifications which will give a complete functional flow of the selected feature to be tested.
- Once the screen prototype and functional specifications are in place, the tester should start writing the cases with the following criteria:
 - UI Tests
 - Functional Cases
 - Database Cases
 - Process Tests



User Name Administrator

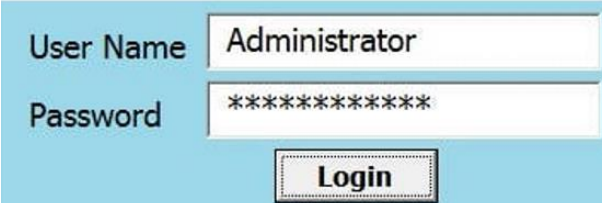
Password *****

Login

[Forgot Password?](#) [Registration](#)

Test Cases Writing Steps – “Login” - Step 3

- UI Tests: The controls/fields which are visible to the user.
 - There are *static control and dynamic controls* available for the feature to be tested.
 - e.g. in the login screen in previous slide, the ‘User Name & Password’ texts are **static fields** which require no user interaction, just for displaying the text only.
- Functional Cases: On the other hand, Login button and the Hyperlinks (Forgot Password? & Registration) are **dynamic fields** which require user interaction by clicking on the controls, which will do some action afterward.



The screenshot shows a login interface on a light blue background. It contains two text input fields: 'User Name' with the value 'Administrator' and 'Password' with masked characters '*****'. Below these is a 'Login' button. At the bottom, there are two hyperlinks: 'Forgot Password?' and 'Registration'.

User Name	Administrator
Password	*****
Login	
Forgot Password?	Registration

Test Cases Writing Steps – “Login” - Step 3- Cont.

- Database Cases: Once the user enters the username and password, the tests may be written to check the related database for, whether the username & password is checked in the right database and table and also the user has the permission to login to the application under test.
- Process Tests: This is related to the process associated with the feature and the functionality.
 - e.g. clicking Forgot Password link in the above sample screen may send an Email to the user. So, maybe an Email needs to be tested for the proper process and confirmation.



A screenshot of a login interface on a light blue background. It features two input fields: 'User Name' with the text 'Administrator' and 'Password' with masked characters '*****'. Below these fields is a 'Login' button with a dotted border. At the bottom, there are two blue underlined links: 'Forgot Password?' and 'Registration'.

User Name	Administrator
Password	*****
Login	
Forgot Password?	Registration

Test Cases Writing Steps – “Login”

- Step 4: Finally, keep the i) Basic Flow ii) Alternative Flow iii) Options and iv) Exceptions for the complete coverage of the functional flow and feature to be tested. Every concept should be applied to positive and negative tests.
 - e.g., let us see the simple approach for the sample login screen in the next slide.

Test Cases Writing Steps – “Login” Example- Step 4

- Basic Flow: Enter the URL path of the Login in any browser and enter the information required and login to the application.
- Alternate Flow: Install the application on a mobile device and enter the information required and login to the application.
- Options: What are the options available to come to the same login screen?
 - For Example, after login to the application, clicking ‘Logout’ may bring the same screen or if the session timeout or session expired, the user may come to the login screen.
- Exceptions: What are exceptions if my tests are negative?
 - For Example, if wrong credentials are entered in the Login screen, whether the user will get an error message or no action associated.

Test Case Document- Writing Steps for login screen

No.	Steps to Reproduce	Expected Behaviour
1.	Open a browser and enter the URL for the Login screen.	The Login screen should be displayed.
2.	Install the app in Android phone and open it.	The Login screen should be displayed.
3.	Open the Login screen and check the texts available are correctly spelled.	'User Name' & 'Password' text should be displayed before the related text box. Login button should have the caption 'Login'. 'Forgot Password?' And 'Registration' should be available as Links.
4.	Enter the text in the User Name box.	Text can be entered by mouse click or focus using tab.
5.	Enter the text in the Password box.	Text can be entered by mouse click or focus using tab.
6.	Click the Forgot Password? Link.	Clicking the link should take the user to the related screen.
7.	Click the Registration Link	Clicking the link should take the user to the related screen.

Test Case Document- Writing Steps for login screen

8.	Enter the user name and password and click the Login button.	Clicking the Login button should take to the related screen or application.
9.	Go to the database and check the correct table name is validated against the input credentials.	The table name should be validated and a status flag should be updated for successful or failure login.
10.	Click the Login without entering any text in the User Name and Password boxes.	Click the Login button should alert a message box 'User Name and Password are Mandatory'.
11.	Click the Login without entering text in the User Name box, but entering text in Password box.	Click the Login button should alert a message box 'Password is Mandatory'.
12.	Click the Login without entering text in the Password box, but entering text in User Name box.	Click the Login button should alert a message box 'User Name is Mandatory'.
13.	Enter the maximum allowed text in the User Name & Password boxes.	Should accept the maximum allowed 30 characters.
14.	Enter the User Name & Password starting with the special characters.	Should not accept the text starting with special characters, which is not allowed in Registration.

Test Case Document- Writing Steps for login screen

15.	Enter the User Name & Password starting with blank spaces.	Should not accept the text stating with blank spaces, which is not allowed in Registration.
16.	Enter the text in the password field.	Should not display the actual text instead should display asterisk * symbol.
17.	Refresh the Login page.	Page should be refreshed with both User Name and Password fields blank.
18.	Enter the User Name.	Depends on the browser auto fill settings, previously entered user names should be displayed as a dropdown.
19.	Enter the Password.	Depends on the browser auto fill settings, previously entered Passwords should NOT be displayed as a dropdown.
20.	Move the focus to Forgot Password link using Tab.	Both mouse click and enter key should be usable.
21.	Move the focus to Registration link using Tab.	Both mouse click and enter key should be usable.
22.	Refresh the Login page and press Enter key.	The Login button should be focussed and the related action should be fired.

Test Case Document- Writing Steps for login screen

23.	Refresh the Login page and press Tab key.	The first focus in the Login screen should be the User Name box.
24.	Enter the User and Password and leave the Login page idle for 10 minutes.	Message box alert 'Session Expired, Enter User Name & Password Again' should be displayed with both User Name & Password fields cleared.
25.	Enter the Login URL in Chrome, Firefox & Internet Explorer browsers.	Same Login screen should be displayed without much deviation on the look and feel and alignment of text and form controls.
26.	Enter the Login credentials and check Login activity in Chrome, Firefox & Internet Explorer browsers.	The action of Login button should be one and the same in all the browsers.
27.	Check the Forgot Password and Registration link is not broken in Chrome, Firefox & Internet Explorer browsers.	Both the links should take to the relative screens in all the browsers.

Test Case Document- Writing Steps for login screen

28.	Check the Login functionality is working properly in Android mobile Phones.	The Login feature should work the same way as it is available in the web version.
29.	Check the Login functionality is working properly in Tab and iPhones.	The Login feature should work the same way as it is available in the web version.
30.	Check the Login screen allows the concurrent users of the system and all the users are getting the Login screen without delays and within the defined time of 5-10 seconds.	This should be achieved using many combination of operating system and browsers either physically or virtually or can be achieved using some performance / load testing tool.

Test Data Collection - login screen

Sl.No.	Purpose of Test Data	Actual Test Data
1.	Test the proper user name and password	Administrator (admin2015)
2.	Test the maximum length of user name and password	Administrator of the Main System (admin2015admin2015admin2015admin)
3.	Test the blank spaces for the user name and password	Enter blank spaces using space key for user name and password
4.	Test the improper user name and password	Admin (Activated) (digx##\$taxk209)
5.	Test the user name and password with uncontrolled spaces between.	Admin istrator (admin 2015)
6.	Test the user name and password starting with special characters	\$%#@#\$Administrator (%##**#admin)

Test Data Collection - login screen

7.	Test the user name and password with all small characters	administrator (admin2015)
8.	Test the user name and password with all capital characters	ADMINISTRATOR (ADMIN2015)
9.	Test the Login with the same user name and password with multiple systems concurrently at the same time.	Administrator (admin2015) - for Chrome in the same machine and different machine with operating system Windows XP, Windows 7, Windows 8 and Windows Server. Administrator (admin2015) - for Firefox in the same machine and different machine with operating system Windows XP, Windows 7, Windows 8 and Windows Server. Administrator (admin2015) - for Internet Explorer in the same machine and different machine with operating system Windows XP, Windows 7, Windows 8 and Windows Server.
10.	Test the Login with the user name and password in the mobile application.	Administrator (admin2015) – for Safari and Opera in Android Mobiles, iPhones and Tablets.